Input Tariff Liberalization and Gender Disparities: Evidence from China

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Abstract

This study examines the impact of input tariff liberalization on gender disparities using China's WTO accession as a natural experiment. We first offer a theoretical framework to illustrate the mechanisms. Using micro-level data from China, we show that input tariff liberalization increases incomes for manufacturing workers but widens the gender income gap. This widening is not due to longer working hours by men but by rising gender wage inequality. We also find that input tariff reductions worsen manufacturing workers' health, and this adverse effect is more pronounced for women. These worsening gender disparities, however, are concentrated primarily among unskilled workers.

Keywords: *input tariff liberalization, gender disparities, health, labor supply elasticity, China*

JEL classification: *F1, I1, J2, J3*

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1 Introduction

How does input tariff liberalization in a large developing country affect men and women differently? Beyond income and wage inequality, does input tariff liberalization also influence other dimensions of gender disparity, such as working time and health? Finally, what role does education play in shaping the effects of input tariff liberalization on gender disparities?

This study contributes to the literature by investigating these important yet underexplored questions. A large number of studies have documented significant distributional effects in developing countries following trade liberalizations (Goldberg and Pavcnik, 2007). However, most existing studies focus on changes in the skill premium, while comparatively little attention has been paid to gender disparities (Juhn et al., 2013, 2014). Moreover, the limited literature that does examine gender outcomes primarily concentrates on income or wage inequality, leaving the effects of trade shocks on other critical dimensions of well-being, such as working time and health, largely unexplored.

To fill these research gaps, we draw on rich micro-level data from China, the world's largest exporting economy and home to the largest manufacturing workforce. Like many developing countries, China has a long history of patriarchal norms. As shown in Table 1, even prior to its accession to the WTO, sizable gender gaps existed in income and hourly wages. Understanding how trade liberalization affects gender disparities in China is therefore of particular importance, with implications for long-term development in China and other developing economies (Duflo, 2012).

China's accession to the WTO also offers a natural experiment for analyzing input tariff shocks. A distinctive feature of China's import structure is its heavy reliance on intermediate inputs, which account for more than 90% of total imports (Fan et al., 2015, 2020).[1]

[1] Final consumption goods account only for about 4% of China's total import value. The rest is intermediate and capital goods, and the latter is also typically considered as intermediate inputs.

Consequently, the tariff reductions implemented following WTO accession primarily affected intermediate inputs. As shown in Figure 1, tariffs on intermediate inputs declined sharply between 1997 and 2015, with the average tariff rate falling from 12.9% in the pre-WTO period to 5.7% in the post-WTO era. This feature allows us to examine an input tariff channel, which complements other channels of trade liberalization, such as export expansion and import competition, explored in the existing literature. Although input tariff liberalization was a nationwide policy and thus exogenous to individual workers and firms, exposure to this shock varied substantially across Chinese prefectures due to differences in initial industrial labor composition (see Figure 2). Our empirical strategy hence takes advantage of this regional variation to identify the effects of input tariff reductions on multiple dimensions of gender disparity.

We begin by documenting a set of stylized facts on gender disparities in income, labor supply, and health outcomes following China's accession to the WTO. After the trade liberalization shock, male earnings—measured by both annual income and hourly wages—increased more than female earnings. In contrast, female workers experienced larger increases in working hours and a greater rise in the likelihood of reporting illness or injury.

To rationalize these observed facts and to guide our empirical analysis, we develop a simple model incorporating imported intermediate inputs and labor market frictions. The model employs a flexible structure allowing for gender-specific heterogeneity on both the labor demand and labor supply sides. A reduction in input tariffs lowers firms' costs of intermediate inputs, leading to an expansion of production and an increase in labor demand. Because men and women may face different demand shifts and exhibit different labor supply responses, their equilibrium outcomes in wages and working hours can differ. Among these mechanisms, differences in labor supply elasticities provide the most plausible explanation for our empirical findings. Consistent with the labor economics literature (e.g.,

Killingsworth and Heckman, 1986; Blundell and Macurdy, 1999; Alesina et al., 2011; Keane, 2011, 2022; McClelland and Mok, 2012; Goussé et al., 2017), we assume that women have a higher labor supply elasticity than men. As a result, following an input tariff reduction, women experience a smaller increase in earnings but a larger increase in working hours. Because both income and working time are key determinants of health (e.g., Hamermesh et al., 2017; Rod et al., 2017; Berniell and Bietenbeck, 2020), this in turn leads to a more pronounced deterioration in health outcomes for women.

We then empirically test the model's predictions. Following Amiti and Konings (2007) and Topalova (2010), we construct a prefecture-level measure of exposure to input tariff reductions using the Chinese province-level input–output table and the initial prefecture-level industry composition of labor. We relate this measure to a range of outcomes capturing gender disparities. We begin by examining income inequality. Using data from both the China Health and Nutrition Survey (CHNS) and the Urban Household Survey (UHS), we find consistent evidence that input tariff liberalization widens the gender income gap. Although incomes increase for both men and women, the gains are significantly smaller for female workers.

Decomposing income into hourly wages and working hours reveals that the widening income gap is not driven by a relative increase in men's working time, but rather by a substantial rise in gender wage inequality. These results are consistent with women having more elastic labor supply than men. As illustrated in Appendix Figure 1, with a flatter labor supply curve, a positive labor demand shock induced by trade liberalization leads to a larger increase in equilibrium hours but a smaller increase in equilibrium wages for female workers.

We next examine health outcomes. Given that lower income and longer working hours can adversely affect health, the model predicts a widening gender health gap. Consistent with this prediction, we find that input tariff liberalization significantly widens gender

disparities in health: female workers are more likely than male workers to report illness or injury in prefectures more exposed to the trade shock. We subject these findings to a wide range of robustness checks, including controlling for additional interaction effects and spousal characteristics, excluding outliers, incorporating prefecture-specific linear trends, employing an instrumental variable (IV) approach, and conducting placebo tests. To further account for labor demand shifts, we also decompose the trade shock into gender-specific components.

Splitting the sample into skilled and unskilled workers based on educational attainment, we find that the widening gender gaps in both income and health are concentrated mainly among unskilled workers. These results suggest that education plays a critical buffering role against the adverse gendered consequences of trade shocks.[2]

We contribute to the relevant literature along several dimensions. First, it speaks to the broad literature on the distributional effects of trade liberalization. Existing studies in this literature have documented that globalization can have a causal effect on income inequality in various countries (e.g., Feenstra and Hanson, 1999; Goldberg and Pavcnik, 2007; Helpman et al., 2010; Han et al., 2012) via different mechanisms (e.g., Goldberg and Pavcnik, 2005; Verhoogen, 2008; Bustos, 2011; Amiti and Davis, 2012; Brambilla et al., 2012).[3] While most of this literature focuses on changes in the skill premium, our study relates more closely to a smaller but growing strand that examines the effects of trade liberalization on gender inequality (e.g., Ozler, 2000; Aguayo-Tellez et al., 2013; Juhn et al., 2013, 2014; Wang et al., 2022). We extend this literature by analyzing a previously unexplored channel—input tariff reductions—and by documenting their effects not only on gender income inequality, but also on other important dimensions of gender disparity,

[2] According to the China Human Capital Report from Center for Human Capital and Labor Market Research, the average years of schooling among the working-age population increased from 6.14 years in 1985 to 10.75 years in 2021 and the gender gap has gradually disappeared (Li, 2023).

[3] Goldberg and Pavcnik (2007) and Pavcnik (2017) provide excellent reviews of this literature.

including working time and health.

Second, our study also contributes to the emerging literature on trade and health. Existing research has primarily examined the health consequences of import competition (e.g., McManus and Schaur, 2016; Pierce and Schott, 2016) or export expansion (e.g., Oster, 2012; Hummels et al., 2016; Bombardini and Li, 2020). In contrast, we focus on the health effects of input tariff liberalization (Fan et al., 2020). Moreover, whereas prior studies largely emphasize average health outcomes, we provide novel evidence on how trade shocks differentially affect the health of male and female workers.

Third, our work complements the literature on the distributional effects of China's WTO accession. Previous studies show that WTO accession was associated with rising wage inequality (Han et al., 2012), that market structure mediates the distributional effects of trade liberalization (Han et al., 2016), and that WTO accession widened income and health gaps between skilled and unskilled manufacturing workers (Fan et al., 2020). While this literature has focused primarily on skill-based inequality, we contribute by providing new evidence on gender disparities.

Finally, our paper is related to the literature on access to imported intermediate inputs. Most existing studies examine firm-level outcomes, such as total factor productivity (e.g., Amiti and Konings, 2007; Kasahara and Rodrigue, 2008; Halpern et al., 2015), product scope (Goldberg et al., 2010), and product quality (e.g., Bas and Strauss-Kahn, 2015; Fan et al., 2015, 2018). We add to this literature by offering new results on gender disparities.

The remainder of the paper is organized as follows. Section 2 describes the policy background and presents stylized facts. Section 3 introduces a simple conceptual framework to guide the empirical analysis. Section 4 outlines the data and empirical strategy. Section 5 presents the main results on gender disparities in income, wages, working time, and health. Section 6 examines the role of education in shaping the gendered effects of input tariff

liberalization. Section 7 concludes.

2 Background and Stylized Facts

To motivate the subsequent analysis, we begin by providing background information and documenting preliminary patterns in gender disparities before and after China's accession to the WTO. As shown in Figure 1, China's import tariffs on intermediate inputs declined sharply between 1997 and 2015, with the key turning point occurring in 2001, when China joined the WTO. Prior to accession, tariffs remained relatively stable; afterward, the average tariff on intermediate inputs fell markedly, from 12.9% in the pre-WTO period to 5.7% in the post-WTO period. This pronounced and policy-driven decline provides an ideal setting for studying the effects of input tariff liberalization.

We next present descriptive evidence on gender-specific outcomes—including income, working hours, and health—before and after WTO accession. Table 1 reports simple mean comparisons by gender across the two periods, with the final column presenting difference-in-differences (DID) estimates. The first two rows report income (in natural logarithms) using the CHNS data and the UHS data, respectively. Details on these data sources are provided in Section 4. Income in the CHNS is expressed in constant 2015 values, while income in the UHS is expressed in constant 2008 values. To mitigate the influence of outliers, all income measures are winsorized at the 1% level in each tail. The results indicate that a substantial gender income gap existed even before WTO accession and widened further afterward. For example, based on the CHNS data, the pre-WTO gender income differential was 0.168 (approximately 872 yuan), equivalent to 12.7% of female workers' average annual income in that period. After WTO accession, the differential increased to 0.305 (approximately 5,340 yuan), corresponding to 33.6% of female workers' average annual income.

Rows (3) and (4) show the results on hourly wage and working time (also in natural logarithms), respectively. A similar pattern emerges for hourly wages: male workers earned higher wages before WTO accession, and the gender wage gap widened in the post-WTO period. In contrast, while women worked fewer hours than men before WTO accession, this pattern reversed afterward, with female working hours increasing relative to those of males. Taken together, these patterns suggest that the widening income gap is driven not by a relative increase in male working time, but by a deterioration in gender wage equality.

Finally, Row (5) compares gender differences in health outcomes before and after WTO accession. Prior to accession, there was no substantial gender disparity in health, and male workers were actually slightly more likely to report illness or injury. Following WTO accession, however, female workers became significantly more likely than male workers to experience illness or injury, indicating a marked widening of the gender health gap.

3 Conceptual Framework

In this section, we develop a simple theoretical framework to rationalize the gender-differentiated responses to input tariff reductions. Intuitively, lower input tariffs reduce the cost of intermediate inputs, thereby inducing an expansion in production. As a result, labor demand increases (Fan et al., 2020). This positive labor demand shock can generate heterogeneous responses in wages and working hours across genders, driven by two broad sources of heterogeneity.

First, labor demand shocks may differ between male and female workers. Such differences can arise from gender-based task specialization within firms. In particular, women may have a comparative disadvantage in “brawn” (physical, blue-collar) tasks, such as production and assembly, and a comparative advantage in “brain” (cognitive, white-collar) tasks, such as marketing and accounting. This results in a gender-based division of labor

(Weinberg, 2000; Rendall, 2010; Juhn et al., 2014). Because these tasks differ in their labor intensity and exposure to imported inputs, male and female labor demand may respond differently to input tariff shocks.

Second, even if labor demand shocks were identical across genders, heterogeneous outcomes could still emerge due to differences in labor supply. A large literature (e.g., Killingsworth and Heckman, 1986; Blundell and Macurdy, 1999; Alesina et al., 2011; Keane, 2011, 2022; McClelland and Mok, 2012; Goussé et al., 2017) documents that female labor supply is substantially more elastic than male labor supply. These differences can be micro-founded through several channels, including endogenous discrimination under monopsonistic employers (Dudek, 2025), home production and intra-household bargaining (Alesina et al., 2011; Goussé et al., 2017), among others

To assess which of these mechanisms can rationalize our empirical findings, we construct a flexible model that allows for both demand-side and supply-side heterogeneity across genders. While the framework admits rich sources of heterogeneity, it ultimately implies that differences in labor supply elasticities play a central role in explaining the observed gender disparities.

3.1 Workers

Let $j \in \{m, f\}$ denote gender, with m for men and f for women. For each gender j, there is a continuum of workers of measure N_j, indexed by $i \in I_j$. For notational simplicity, we will introduce individual indices only when necessary and omit them when no confusion is likely to arise. Conditional on employment, a worker's problem is to maximize

$$u_j = c - g_j(a, \ell), \tag{1}$$

where c is consumption, a is illness, and ℓ is working hours. The function $g_j(\cdot)$ captures disutility from poor health and long working hours. As established in both the

medical and economics literatures (Berniell and Bietenbeck, 2020; Hummels et al., 2025), heavy workloads are a major source of adverse health outcomes. Accordingly, we model illness as a function of hours worked and specify

$$g_j(a,\ell) = g_j\big(q_j(\ell),\ell\big) = \xi_j \frac{\ell^{1+\frac{1}{\varepsilon_j}}}{1+\frac{1}{\varepsilon_j}}, \tag{2}$$

where $a = q_j(\ell)$ and $\varepsilon_j > 0$. Imposing $c = w_j\ell$, the optimal labor supply given hourly wage w_j is

$$\ell = \xi_j^{-\varepsilon_j} w_j^{\varepsilon_j} = \tilde{\xi}_j w_j^{\varepsilon_j}. \tag{3}$$

The utility derived from employment is

$$u_j(w_j) = \frac{\tilde{\xi}_j}{\varepsilon_j+1} w_j^{\varepsilon_j+1}. \tag{4}$$

3.2 Firms

There is a continuum of final-goods firms of measure one, indexed by $\omega \in \Omega$. Each firm purchases a bundle of intermediate inputs M and combines it with two types of labor, male and female to produce output according to

$$y = \phi F(L_m, L_f, M)^{\alpha}, \tag{5}$$

where

$$F(L_m, L_f, M) = L_m^{\beta_m} L_f^{\beta_f} M^{\beta_M}, \tag{6}$$

$$L_j = n_j \ell_j, \tag{7}$$

$$M = \left(\int_0^1 m(s)^{\frac{\sigma-1}{\sigma}} ds\right)^{\frac{\sigma}{\sigma-1}}, \tag{8}$$

In line with Lucas's (1978) span-of-control model, final-good production exhibits decreasing returns to scale, with $\beta_m + \beta_f + \beta_M = 1$ and $\alpha \in (0,1)$. n_j is the number of employed workers with gender j, and ℓ_j is their working hours. M is bundled from a continuum of heterogenous input varieties $s \in [0,1]$.

Markets for intermediate inputs and final goods are competitive, and the final-good price is normalized to 1. Let P_M denote the price index of M. The optimal input sourcing decision implies

$$M = \frac{\alpha\beta_M}{P_M} y, \tag{9}$$

$$y = \phi^{\frac{1}{1-\alpha\beta_M}} \left(\frac{\alpha\beta_M}{P_M}\right)^{\frac{\alpha\beta_M}{1-\alpha\beta_M}} L_m^{\frac{\alpha\beta_m}{1-\alpha\beta_M}} L_f^{\frac{\alpha\beta_f}{1-\alpha\beta_M}} = \Phi L_m^{\tilde{\beta}_m} L_f^{\tilde{\beta}_f}, \tag{10}$$

where $\tilde{\beta}_m + \tilde{\beta}_f < 1$.

The labor market is imperfect due to matching frictions between firms and workers, which limit workers' ability to move freely across firms. We model these frictions using a discrete-choice framework (Setzler and Tintelnot, 2021). To attract workers with gender j, a firm ω posts $t_j(\omega)$ job vacancies with a promised hourly wage $w_j(\omega)$. The cost of each posted vacancy is wage-dependent and is denoted by $h_j(w_j(\omega))$. Observing $\int_{\omega\in\Omega} t_j(\omega)d\omega$ posted vacancies in total, each worker i chooses one job offer among them. The total utility of accepting job offer $k(\omega) \in [0, t_j(\omega)]$ is $u_j(w_j(\omega))\zeta(i, k(\omega))$, where $u_j(w_j(\omega))$ is the wage-dependent component defined in equation (4), and $\zeta(i, k(\omega))$ is a post-specific idiosyncratic utility shock. We assume that the vector of $\zeta(i, k(\omega))$ across all posts follows a multivariate Fréchet distribution with shape parameter $\theta_j > 0$ and correlation coefficient $\rho_j \in (0,1)$ within each nest ω.[4] This yields a nested logit structure. The probability that worker i chooses any post from firm ω is

$$\Pr(i, \omega) = \frac{t_j(\omega)^{\frac{1}{1-\rho_j}} u_j(w_j(\omega))^{\theta_j}}{\int_{w\in\Omega} t_j(\omega\prime)^{\frac{1}{1-\rho_j}} u_j(w_j(\omega\prime))^{\theta_j}} = \frac{t_j(\omega)^{\frac{1}{1-\rho_j}} u_j(w_j(\omega))^{\theta_j}}{\bar{U}_j}, \tag{11}$$

where $\bar{U}_j$ is exogenous from the perspective of an individual firm.

[4] The CDF is $\Pr\{\ldots, \zeta(i, k(\omega)) \le \bar{\zeta}(i, k(\omega)), \ldots, \zeta(i, k'(\omega')) \le \bar{\zeta}(i, k'(\omega')), \ldots\} = \exp\left\{-\int_{\omega\in\Omega}\left(\int_0^{t_j(\omega)} \bar{\zeta}(i, k(\omega))^{-\frac{\theta}{1-\rho_j}}\right)^{1-\theta_j}\right\}$.

The distribution of $\zeta(i, k(\omega))$ is identical across all workers, so the total number of workers of gender j hired by a firm equals $n_j = \frac{N_j}{\bar{U}_j} t_j^{1-\rho_j} u_j(w_j)^{\theta}$. For simplicity, we parameterize the total job-posting cost $h_j(w_j)t_j$ as

$$C_j(n_j, w_j) = \frac{\bar{U}_j}{N_j} n_j^{\frac{1}{1-\rho_j}} h_j(w_j) u_j(w_j)^{-\frac{\theta}{1-\rho_j}} = A_j n_j^{\delta_j} w_j^{\lambda_j}, \tag{12}$$

where $\delta_j = \frac{1}{1-\rho_j} > 1$. We assume $0 < \lambda_j < \delta_j$ considering that $u_j(w_j)^{-\frac{\theta}{1-\rho_j}}$ reduces the convexity of posting cost regarding wages.

Therefore, the profit of a firm is

$$\pi = y - \sum_{j\in\{m,f\}} w_j n_j \ell_j - \sum_{j\in\{m,f\}} C_j(n_j, w_j). \tag{13}$$

The firm has monopsony power over the wage levels. Substituting equations (3), (10), and (12) into (13), the first-order conditions regarding n_j and w_j are

$$\tilde{\beta}_j \frac{y}{n_j} = \frac{w_j n_j \ell_j}{n_j} + \delta_j \frac{C_j(n_j, w_j)}{n_j}, \tag{14}$$

$$\varepsilon_j \tilde{\beta}_j \frac{y}{w_j} = (\varepsilon_j + 1) \frac{w_j n_j \ell_j}{w_j} + \lambda_j \frac{C_j(n_j, w_j)}{w_j}. \tag{15}$$

Combining these two conditions yields

$$n_j = \left(\frac{\tilde{\xi}_j}{\varepsilon_j \delta_j - \lambda_j}\right)^{\frac{1}{\delta_j+1}} w_j^{\frac{\varepsilon_j+1-\lambda_j}{\delta_j+1}} = B_j w_j^{\frac{\varepsilon_j+1-\lambda_j}{\delta_j+1}}. \tag{16}$$

Plugging n_j into equation (14), we have

$$\tilde{\beta}_j \Phi \left(B_m \tilde{\xi}_m w_m^{\frac{\delta_m}{\delta_m-1}\varepsilon_m + \frac{\delta_m-\lambda_m}{\delta_m-1}}\right)^{\tilde{\beta}_m} \left(B_f \tilde{\xi}_f w_f^{\frac{\delta_f}{\delta_f-1}\varepsilon_f + \frac{\delta_f-\lambda_f}{\delta_f-1}}\right)^{\tilde{\beta}_f}$$

$$= (B_j \tilde{\xi}_j + \delta_j A_j B_j^{\delta_j}) w_j^{\frac{\delta_j}{\delta_j-1}\varepsilon_j + \frac{\delta_j-\lambda_j}{\delta_j-1}}, \tag{17}$$

for each $j \in \{m, f\}$.

3.3 Input tariffs

Input tariffs are incorporated using a standard Ricardian trade framework (Eaton and

Kortum, 2002; Antràs et al., 2017). The final-goods firm can source each input variety from both domestic (D) and foreign (F) upstream producers. Markets for each variety s are competitive, with factory-gate prices denoted by $p_D(s)$ for domestic producers and $p_F(s)$ for foreign producers. We do not model the production process in the upstream sector explicitly and instead treat these factory-gate prices as exogenous. The corresponding purchaser prices are $p_D(s)$ for domestic inputs and $\tau p_F(s)$ for imported inputs, where $\tau > 1$ is one plus the tariff imposed on imported inputs.

For each input variety, the final-goods firm chooses the source with the lowest cost: $p_M(s) = \min\{p_D(s), \tau p_F(s)\}$. We assume that the inverse of factory-gate prices follows a Fréchet distribution:

$$\Pr(p_o(\omega) \geq p) = \exp(-T_o p^{\kappa}), \text{ for } o \in \{D, F\}, \tag{18}$$

where $\kappa > \sigma - 1$. Therefore, the price index for the input bundle M would be

$$P_M^{-\kappa} = \Gamma\left(\frac{\kappa+1-\sigma}{\kappa}\right)^{\frac{\kappa}{1-\sigma}} (T_D + T_F \tau^{-\kappa}), \tag{19}$$

where Γ is the gamma function.

3.4 Trade liberalization

Now we consider a decrease in the import tariff for intermediate inputs, i.e., $d\ln\tau < 0$. From equations (19) and (10) we can see that

$$d\ln P_M = \chi_F d\ln\tau < 0, \tag{20}$$

$$d\ln\Phi = -(1 - \alpha\beta_M) d\ln P_M > 0, \tag{21}$$

where $\chi_F = \frac{T_F \tau^{-\kappa}}{T_D + T_F \tau^{-\kappa}}$ is the initial import share for intermediate inputs. Therefore, the tariff cut decreases the cost for the input bundle and in turn increases labor productivity.

Next, we log-linearize equation (17) and get the following system of equations:

$$(1 - \tilde{\beta}_m) H_m d\ln w_m - \tilde{\beta}_f H_f d\ln w_f = d\ln\Phi, \tag{22}$$

$$-\tilde{\beta}_m H_m d\ln w_m + (1 - \tilde{\beta}_f) H_f d\ln w_f = d\ln\Phi, \tag{23}$$

where $H_j = \frac{\delta_j}{\delta_j - 1}\varepsilon_j + \frac{\delta_j - \lambda_j}{\delta_j - 1}$. The solution yields

$$d\ln w_m = \frac{d\ln\Phi}{(1-\widetilde{\beta}_m - \widetilde{\beta}_f)H_m} > 0, \tag{24}$$

$$d\ln w_f = \frac{d\ln\Phi}{(1-\widetilde{\beta}_m - \widetilde{\beta}_f)H_f} > 0. \tag{25}$$

The hourly wages for both genders would rise following the tariff shock. Moreover, since $d\ln\ell_j = \varepsilon_j d\ln w_j$ and $d\ln w_j\ell_j = (\varepsilon_j + 1)d\ln w_j$ for both $j \in \{m, f\}$ based on equation (3), the working hours and total earnings of each worker would also increase. We have the following proposition:

Proposition 1: *The total earnings, hourly wages, and working hours for both genders increase after the reduction in input tariffs.*

3.5 Gender disparities

In this flexible framework, we allow for gender differences in multiple dimensions, such as labor supply elasticity (ε_j), input intensity (β_j), and the parameters for hiring costs (δ_j and λ_j). These differences can generate rich patterns of gender disparities. Here, however, we focus on the main empirical findings: (i) women's earnings increase less than men's; (ii) women's working hours increase more than men's; and (iii) women's health deteriorates more than men's. To account for this set of patterns, differences in labor supply elasticity ε_j provide the most plausible explanation, which is also consistent with a large body of literature suggesting that female labor supply elasticities are higher than those of males.

Define a function $q(\varepsilon) = \frac{1}{H} = \frac{1}{\frac{\delta}{\delta-1}\varepsilon + \frac{\delta-\lambda}{\delta-1}}$, we have $\frac{\partial q(\varepsilon)}{\partial \varepsilon} = \frac{\frac{-\delta}{\delta-1}}{H^2} < 0$, $\frac{\partial \varepsilon q(\varepsilon)}{\partial \varepsilon} = \frac{\frac{\delta-\lambda}{\delta-1}}{H^2} > 0$ and $\frac{\partial(\varepsilon+1)q(\varepsilon)}{\partial \varepsilon} = \frac{\frac{-\lambda}{\delta-1}}{H^2} < 0$. Given $\varepsilon_f > \varepsilon_m$, and abstracting from any other parameter differences across genders, we obtain

$$\frac{d\ln w_f}{d\ln w_m} = \frac{H_m}{H_f} < 1, \tag{26}$$

$$\frac{d\ln \ell_f}{d\ln \ell_m} = \frac{\varepsilon_f}{H_f} / \frac{\varepsilon_m}{H_m} > 1, \tag{27}$$

$$\frac{d\ln w_f \ell_f}{d\ln w_m \ell_m} = \frac{\varepsilon_f + 1}{H_f} / \frac{\varepsilon_m + 1}{H_m} < 1. \tag{28}$$

This leads to the following propositions:

Proposition 2: *The total earnings and hourly wages for female workers increase less than those for male workers after the reduction in input tariffs.*

Proposition 3: *The working hours for female workers increase more than those for male workers after the reduction in input tariffs.*

Finally, given that illness is a function of working hours: $a = q_j(\ell)$. Longer hours increase workload and reduce the worker's resistance to illness. We assume that $a \propto \ln\ell$, which yields the following proposition:

Proposition 4: *The probability of illness for female workers increases more than that for male workers after the reduction in input tariffs.*

Finally, to explain why alternative parameterizations—specifically, those introducing heterogeneity only to the demand side—fail to replicate our full set of empirical observations, we argue that the inverse relationship between the gender gaps in hourly wages and working hours is crucial. Empirically, we observe $\frac{d\ln w_f}{d\ln w_m} < 1$ and $\frac{d\ln \ell_f}{d\ln \ell_m} > 1$. Theoretically, given that workers are price takers, the following must hold: $\frac{d\ln \ell_f}{d\ln \ell_m} = \frac{\varepsilon_f}{\varepsilon_m}\frac{d\ln w_f}{d\ln w_m}$. Therefore, reconciling the empirical facts necessarily requires $\varepsilon_f > \varepsilon_m$. For instance, consider a scenario with homogeneous labor supply elasticities ($\varepsilon_f = \varepsilon_m$) but differing hiring costs ($\lambda_f < \lambda_m$). While this could generate a wage gap $\frac{d\ln w_f}{d\ln w_m} = \frac{H_m}{H_f} < 1$, it would also imply $\frac{d\ln \ell_f}{d\ln \ell_m} < 1$, which contradicts the empirical finding on working hours.

4 Data and Empirical Strategy

4.1 Data sources

Our study employs two main individual-level survey datasets, including the CHNS and the UHS. CHNS is an international collaborative project by the Carolina Population Center at the University of North Carolina at Chapel Hill and the National Institute for Nutrition and Health (NINH) at the Chinese Center for Disease Control and Prevention (CCDC). This survey uses a multistage, random cluster process to draw samples. It is conducted once every 2 or 3 years with a total of 10 waves, including 1989, 1991, 1993, 1997, 2000, 2004, 2006, 2009, 2011, and 2015. The data is mainly pooled cross-section in nature.

This survey contains detailed individual characteristics, including age, gender, education, time allocation, wage, occupation, and employer ownership status. It also contains rich information on individual health conditions, including illness and injury status, disease history, smoking behavior, and health insurance coverage. Due to severe missing data in the early waves, we restrict our analysis to waves from 1997 onward.

Since the effects of input tariff shocks are concentrated in the manufacturing sector (Fan et al., 2020), we focus on manufacturing workers in the CHNS data.[5] We further restrict the sample to individuals aged 16 (the minimum legal working age in China) to 60 for men and 55 for women (the retirement ages for workers in China). Finally, to isolate the effect of informal jobs, we also exclude individuals whose weekly working hours are less than 10. Part A of the Data Appendix in the Online Appendix describes the details of the sample selection criteria. The resulting CHNS sample used in our main analysis includes approximately 7,000 manufacturing workers. For placebo tests, we also use workers in the agricultural and service sectors.

[5] Individuals whose occupations are coded as 6 or 7 (see Panel A of Appendix Table 1 for the full industries code).

A limitation of the CHNS is its relatively narrow geographic coverage: it spans only 52 prefectures across 11 provinces, resulting in a modest sample size. To assess the robustness of our findings, we therefore complement the CHNS with the UHS. The UHS is the official source of living standard indicators for urban households in China and is administered by the National Bureau of Statistics. Like the CHNS, it is primarily pooled cross-section, and its key advantage lies in its extensive coverage: 242 prefecture-level cities across all 31 provinces. The UHS has two main limitations. First, it reports only annual income and does not contain information on hourly wages, working hours, or health outcomes, restricting its use to income-based analyses. Second, it includes fewer individual characteristics than the CHNS, covering only basic demographics (gender, age, education) and employment information (working status, occupation, and sector).

In the UHS sample, we again focus on manufacturing workers aged 16 to 55 for women and 60 for men.[6] Detailed sample selection criteria are provided in Part B of the Data Appendix. The final UHS sample consists of approximately 82,000 manufacturing workers observed between 1999 and 2008.

4.2 Measure of input tariff shock

We construct a prefecture-level measure of input tariff exposure following the approach of Amiti and Konings (2007) and Fan et al. (2020). First, we use the SITC classification as an intermediary to map HS product codes to input–output (IO) industry codes. We then compute a sector-level output tariff, $\tau_{k,t}^{output}$, defined as China's import tariff on output of sector k in year t. This measure is calculated as an import-value-weighted average of China's effectively applied HS 6-digit import tariffs (AHS), obtained from the World Integrated Trade Solution (WITS) database, within each two-digit IO industry. Next, we use China's 2002 province-level input-output table to construct the weighted input tariffs

[6] Individuals whose occupations are coded as 3 (see Panel B of Appendix Table 1 for the full industries code).

at the provincial level. Specifically, we compute the province-sector input tariffs as follows:

$$\tau_{i,s,t}^{input} = \sum_k a_{i,s,k} \tau_{k,t}^{output}, \tag{29}$$

where $a_{i,s,k}$ is the share of sector s's total costs spent on inputs supplied by industry k in province i.

Finally, we aggregate these sector-level measures to the prefecture level using employment weights from the Annual Survey of Industrial Firms (ASIF). The prefecture-level input tariff exposure is constructed as

$$tariff_{p,t} = \frac{\sum_s Worker_{p,s,0} \tau_{i,s,t}^{input}}{Total\ Worker_{p,0}}, \tag{30}$$

where $Worker_{p,s,0}$ is the number of workers in sector s and prefecture p at the beginning of our sample period, and $Total\ Worker_{p,0}$ is the total number of workers in prefecture p at that time. The two numbers are computed using firm-level data from the ASIF in China. By construction, reductions in input tariffs correspond to declines in this measure.

Substantial cross-prefecture variation in sectoral employment shares implies considerable heterogeneity in exposure to tariff shocks. Figure 2 shows the regional exposures to the input tariff change over the period of 1997-2007. It is evident that there are large variations in the exposure to the trade shock across Chinese prefectures.

To further assess whether our results are driven by gender-specific labor demand shocks, we follow the decomposition approach of Autor et al. (2019). Specifically, we split the prefecture-level tariff measure into components based on initial female and male employment shares within each industry:

$$tariff_{p,t}^{female} = \frac{\sum_s f_{p,s,0} Worker_{p,s,0} \tau_{i,s,t}^{input}}{Total\ Worker_{p,0}} \tag{31}$$

$$tariff_{p,t}^{male} = \frac{\sum_s (1-f_{p,s,0}) Worker_{p,s,0} \tau_{i,s,t}^{input}}{Total\ Worker_{p,0}}, \tag{32}$$

where $f_{p,s,0}$ denotes the initial share of female workers in industry s and prefecture p. $tariff_{p,t}^{female}$ captures the trade-induced demand shocks specifically affecting female-intensive industries, while $tariff_{p,t}^{male}$ reflects the shocks to male-intensive sectors.

4.3 Outcome variables and control variables

We examine several key labor market outcomes, including individuals' annual income, hourly wage, working time, and health. To study income effects, we use the natural logarithm of annual labor income for manufacturing workers, drawing on both the CHNS and the UHS. Analyses of hourly wages and working hours rely exclusively on the CHNS, which reports hourly earnings and weekly hours worked; both variables are log-transformed. Health outcomes are also examined using the CHNS. Our primary health measure is a binary indicator equal to one if the individual reports having experienced an illness or injury in the past four weeks and zero otherwise.

We include two sets of control variables: individual-level and prefecture-level. Individual-level controls are drawn from the CHNS and UHS. Prefecture-level variables are compiled from national, provincial, and prefectural statistical yearbooks, the WITS database, and census data. Variable definitions and data sources are reported in Online Appendix Table 2, while summary statistics for the CHNS and UHS samples appear in Online Appendix Tables 3 and 4, respectively.

4.4 Empirical strategy

We employ the following benchmark empirical specification to examine the effects of input tariff reductions on a variety of measures of gender disparity in China:

$$y_{it} = \alpha Tariff_{p,t-1} + \beta Tariff_{p,t-1} * Female_i + \gamma X_{it} + \delta X_{p,t-1} + \lambda_t + \lambda_p + \epsilon_{it} \quad (33)$$

where i and t represent the individual and year, respectively. y_{it} is a measure of individual i's outcome, such as income, wage, health, etc. $Tariff_{p,t-1}$ is the lagged prefecture-level input tariff exposure, and its interaction with the female indicator,

$Tariff_{p,t-1} * Female_i$, is our main variable of interest. The coefficient β captures the differential impact of trade liberalization on female relative to male workers.

X_{it} is a comprehensive set of individual-level controls. In income, wage, and hours regressions, these controls include gender, education, age and age squared, residential status, marital status, minority status, occupation dummies, and employer ownership dummies. In health regressions, we additionally control for disease history, smoking behavior, health insurance coverage, and a binary indicator for abnormal BMI.

We also include a set of prefecture-level controls, $X_{p,t-1}$, in our regressions. Specifically, we control for other trade-related policy shocks, including output tariff shocks ($Tariff_Output$), the eliminations of trade uncertainty (e.g., Pierce and Schott, 2016; Handley and Limão, 2017) and export licenses (Bai et al., 2017), and the changes of quotas due to the expiration of global MFA.[7] In health regressions, we also include $Air\ Quality\ Index$ in $X_{p,t-1}$, a prefecture-level measure of air quality to capture the effect of air pollution on health. We also include year fixed effects, λ_t, and prefecture fixed effects, λ_p, in all regressions to account for yearly shocks common to all individuals and all time-invariant differences across prefectures.

5 Main Results

5.1 The effect on income

We begin by examining how input tariff shocks affect manufacturing workers' income. We estimate Equation (33) using the natural logarithm of annual income as the dependent variable, and report the results in the first two columns of Table 2. Column (1) presents estimates based on the CHNS sample. We regress income on the input tariff shock, a female

[7] The output tariff shock is constructed as $Tariff_Output_{p,t} = \frac{\sum_s Worker_{p,s,0}\tau_{k,t}^{output}}{Total\ Worker_{p,0}}$.

dummy, and their interaction, controlling for individual characteristics, a set of prefecture-level trade policy variables, and prefecture and year fixed effects.

The coefficient on the input tariff shock is negative and statistically significant, indicating that reductions in input tariffs (a fall in $Tariff$) significantly increase the income of male manufacturing workers. However, the coefficient on our variable of main interest, $Tariff * Female$, is significantly positive. This implies that female workers benefit significantly less from input tariff reductions than male workers, leading to a widening gender income gap. The estimated interaction effect is also quantitatively sizable: a one-standard-deviation reduction in our prefecture-level input tariff is associated with an increase in the gender income gap of 680 yuan, measured in constant 2015 value.[8] Since the coefficient on the interaction term is smaller than that on the input tariff variable, the overall result implies that, in absolute terms, input tariff reductions still increase female workers' income.

As for control variables, we find that more years of education or being married is associated with a significantly higher income. The coefficient on age is significantly positive but that on the age square is significantly negative. We also find that output tariff reductions, the elimination of policy uncertainty and the removal of quotas significantly increase the income in the manufacturing sector. Other controls are statistically insignificant.

A potential limitation of the CHNS is its relatively narrow geographic coverage, spanning only 52 prefectures in 11 provinces. To assess robustness, we replicate the analysis using the UHS, which covers 242 prefectures across all 31 provinces. As discussed in Section 2.1, the UHS contains more limited individual information, restricting us to controls for age, age squared, education, occupation type, and employer ownership.

Column (2) of Table 2 reports the UHS estimates. The results closely mirror those from

[8] This effect is computed as [exp(0.852*0.06)-1]*exp(9.470)≈680, where 0.06 is the standard deviation of the input tariff shock variable.

the CHNS. Input tariff reductions raise income for all workers, but the gains are significantly smaller for women, resulting in higher gender income inequality. Quantitatively, a one-standard-deviation reduction in the input tariff shock increases the gender income gap by 368 yuan in constant 2008 prices.[9] The consistency across datasets strengthens the robustness of our findings.

5.2 The effects on hourly wage and working time

To understand the drivers underlying the increase in gender income inequality, we next examine hourly wages and working time separately. Columns (3) and (4) of Table 2 report estimates from Equation (33) using log hourly wages and log weekly working hours as dependent variables, respectively. Column (3) shows that the coefficient on the input tariff shock is negative and statistically significant, indicating that tariff reductions raise hourly wages for male workers. The interaction term with the female dummy is positive and significant, but smaller in magnitude. Together, these estimates imply that wage gains from input tariff reductions accrue to both men and women, but are significantly larger for men, leading to a widening gender wage gap. Column (4) focuses on working time. The coefficient on the input tariff shock is negative but statistically insignificant, suggesting that tariff reductions do not significantly affect male workers' hours. In contrast, the interaction term is negative and statistically significant, indicating that female workers experience a relative increase in working time following tariff reductions.

Taken together, the results in Columns (3) and (4) suggest that the worsening of gender income inequality documented earlier is driven by longer working hours of male workers. Instead, it reflects a widening gender wage gap. Although female workers increase their working time relative to men, their relative income nevertheless declines.

[9] Since the UHS data does not provide corresponding CPI data for different regions like CHNS, we adjust the income in UHS data by the CPI in national level to the last wave (2008). [exp(1.035*0.036)-1]*exp(9.179) ≈ 368, where 0.036 is the standard deviation of the input tariff shock variable.

5.3 The effect on health

The results in Table 2 indicate that input tariff liberalization reduces female workers' relative income while increasing their relative working time. Both lower income and longer working hours are likely to adversely affect health, potentially widening gender health disparities. We therefore turn to examining the health consequences of input tariff reductions.

Table 3 reports the health results. In Column (1), we estimate Equation (33) using a binary indicator for whether an individual experienced illness or injury in the past four weeks as the dependent variable. Besides the baseline controls, we also add individual health-related variables, including disease history, smoking behavior, abnormal BMI, and health insurance coverage, as well as a prefecture-level Air Quality Index as additional controls.

We find that input tariff reductions also result in gender health inequality, as the coefficient on the interaction term between the input tariff shock measure and the female dummy is significantly negative. The magnitude is substantial: a one-standard-deviation reduction in our input tariff shock measure is associated with a 2.4-percentage-point widening of the gender gap in the probability of experiencing illness or injury.[10] This effect is large relative to the sample mean incidence rate of less than 9 percent.

To ensure our results are not driven by the choice of the linear probability model, Column (2) reports estimates from a Logit specification. Evaluated at the sample mean, a one-standard-deviation reduction in the tariff shock increases the gender health gap by approximately 2.7 percentage points.[11] The close correspondence between the two specifications confirms the robustness of the result.

To further explore the mechanisms, Columns (3) and (4) add individual income and labor burden as controls. Labor burden here is a comprehensive measure that includes both

[10] $[\exp(0.359*0.06)-1]*\exp(0.087) \approx 0.0238$
[11] $5.683*0.087*(1-0.087)*0.06 \approx 0.0271$

paid work and time spent on domestic chores.[12] Consistent with the medical and labor economics literature, both lower income and higher labor burden are indeed key determinants of poorer health. The coefficients on these two variables are both significant with expected signs. In Columns (5) and (6), we additionally interact income and labor burden with the female dummy. The interaction terms are statistically insignificant, indicating no gender differences in health with respect to income or labor burden. The coefficient of the interaction term between the tariff shock and the female dummy, however, remains negative and statistically significant in all regressions. This suggests that input tariff reductions can affect gender health gaps also through other mechanisms that go beyond the income and labor burden channels.

5.4 Robustness checks

In this subsection, we conduct a battery of sensitivity analyses to ensure the robustness of our main findings. We first check if our results are sensitive to the inclusion of the interaction effects between input tariff and other individual characteristics. For example, if female workers are relatively younger and have lower educational attainment than male workers on average, then our results may reflect the difference in age or education rather than gender. To address this concern, we interact the tariff shock with age, age squared, and years of education, and include these interactions as additional controls. Table 4 shows that the main results remain unchanged.

Second, given that 70–80 percent of individuals in our sample are married, gender differences in health outcomes could potentially reflect intra-household dynamics or an "added worker effect" (Lundberg, 1985). For instance, a spouse's labor market status or education level might influence an individual's health through shared household income or psychological support (Huang et al., 2009; Guo et al., 2020). To account for this, we restrict

[12] The results are similar if we only consider formal working hours instead of the total labor burden.

the sample to married individuals and control for spouse characteristics, including age, education, and employment status. As shown in Table 5, the interaction between the tariff shock and the female dummy remains statistically significant, indicating a gendered effect of trade liberalization that is not mediated by spousal characteristics.

Next, to address the concern on the exogeneity of input tariff changes, we follow the existing literature (e.g., Goldberg and Pavcnik, 2005; Kovak, 2013; Fan et al., 2020) and examine the relationship between the weighted average input tariff changes and weighted average initial input tariffs across prefectures. A negative relationship between the two would indicate that the reductions were determined mainly by the pre-liberalization tariff levels rather than political economy considerations. To check if this is the case, we plot a prefecture's change in input tariff from 1997 to 2007 against its pre-WTO tariff rates in 1997 in Figure 3. Both for all prefectures (Panel A) and for those in the CHNS sample (Panel B), prefectures with higher initial tariffs experienced larger reductions, consistent with WTO-driven liberalization rather than endogenous policy selection. Only one prefecture city, Shiyan, seems to deviate from this negative linear relationship. We hence conduct an additional robustness check by excluding this outlier prefecture from our CHNS sample. As shown in Panel A of Table 6, our main findings still hold after dropping this outlier.

Another potential concern of our results is whether they are sensitive to controlling for prefecture-specific linear trends. Panel B of Table 6 shows that including these trends does not affect our main findings. Next, we also follow Erten and Leight (2021) and Fan et al. (2020) to conduct a test for pre-trends by evaluating the correlations between initial tariffs and outcomes in different years. Specifically, we replace the interaction term with a series of interaction terms between the initial tariff shock in 1997 and the survey wave dummies. Since the initial tariff shock has no time variation, its interaction with the 1997 dummy is omitted due to including prefecture fixed effects in the regression. The estimated

coefficients for the rest of the interaction terms are demonstrated graphically in Figure 4. The post-WTO coefficients are consistently significant and large in magnitude. The graphical patterns are thus consistent with the view that initial tariffs are uncorrelated with pre-WTO outcomes, but predictive of those in the post-WTO period.

While we construct the input tariff shock measure using the labor share at the beginning of the sample as weights, endogeneity could still be a concern. Additionally, as the labor weights are derived from data on large-scale firms, which may not perfectly represent the employment structure of the entire manufacturing sector within each prefecture. To address these endogeneity and measurement concerns, we adopt an IV strategy that isolates the exogenous policy variation from the pre-WTO period.

Specifically, we instrument the tariff shock and its interaction with gender using the 1997 initial tariff level interacted with wave dummies: $Tariff_{p,1997} * Wave_t$ and $Tariff_{p,1997} * Female * Wave_t$. The 1997 tariff level represents the protection status prior to China's major WTO commitments. The relevance of this instrument stems from the fact that industries with higher initial protection were required to implement larger tariff reductions to comply with WTO accession agreements (Brandt et al., 2017). Moreover, the tariff levels set in 1997—determined years before our sample period—are unlikely to affect individual outcomes in later years through channels other than the subsequent trade liberalization path.

Table 7 presents the results from the IV regressions. Across all specifications, the interaction between the tariff shock and the female dummy remains statistically significant. The validity of our IV strategy hinges on the relevance of the pre-WTO tariff structure. As visually suggested in Figure 3, a prefecture's initial tariff rate in 1997 is negatively correlated with its subsequent tariff changes from 1997 to 2007, reflecting the "catch-up" liberalization required by WTO commitments. We also report the Kleibergen-Paap rk Wald

F-statistic at the bottom of Table 7 to assess the strength of our instruments. The F-statistic exceeds the conventional threshold, allowing us to reject the null hypothesis of weak instruments.[13]

Finally, we also perform a set of placebo tests to rule out the possibility that our findings are driven by some prefecture-level shocks that influence all individuals. If the effects on wages, health conditions, and working time are indeed caused by input tariff shocks, they should affect mainly workers in the manufacturing sector. Table 8 shows the placebo test results. Panel A shows no significant effects for workers in agriculture.[14] Panel B shows that, in the service sector, input tariff reductions have no differential effects on working time or health, and in fact increase female workers' income and wages relative to men. These findings reinforce the interpretation that our results operate primarily through manufacturing-sector exposure to input tariff shocks.

5.5 Accounting for demand heterogeneity

A potential alternative explanation is that trade liberalization disproportionately affected female-intensive industries. If this were the case, the gender health gap would be a simple reflection of differential labor demand shocks rather than differences in labor supply behavior. To evaluate this possibility, we replace the aggregate tariff shock with gender-specific shocks based on initial employment shares: $tariff_{p,t}^{female}$ and $tariff_{p,t}^{male}$. Table 9 shows that the estimated effects of these two components on income, wages, working hours, and health are remarkably similar and closely aligned with our baseline results. The absence of meaningful differences between female- and male-specific shocks suggests that

[13] We present a simplified first-stage analysis at the prefecture-year level in Appendix Table 5. We regress the current tariff levels on the interactions between the 1997 initial tariffs and wave dummies. Across all samples, the coefficients on the instruments are consistently negative and statistically significant. More importantly, the absolute values of these coefficients increase in later waves, confirming that industries with higher initial protection experienced progressively larger tariff reductions over time. This systematic pattern provides strong empirical support for the relevance of our instrument set.

[14] Since quite a few individuals in the agriculture sector do not have a formal wage like those in the manufacturing or service sector, we define their yearly income based on their income from farming, fishing, and raising livestock, and calculate their hourly wage by dividing their incomes by their working time.

the widening gender gaps are not primarily driven by differential industry exposure. Instead, the evidence points to differences in labor supply responses as the dominant mechanism underlying our findings.

6 The Role of Education

Existing studies in the literature emphasize education as a key determinant of the distributional consequences of trade liberalization. Motivated by this insight, we examine whether education mediates the impact of input tariff reductions on gender disparities in income, labor supply, and health. We split the CHNS manufacturing sample into two groups based on educational attainment. Workers with at least a high school diploma are classified as *skilled*, while those without a high school education are classified as *unskilled*. We then re-estimate our baseline specifications separately for these two subsamples.

Table 10 reports the subsample regression results. Columns (1) and (2) present estimates for annual income. The results indicate that the increase in gender income inequality documented earlier is driven almost entirely by the unskilled group. For skilled workers, neither the coefficient on the input tariff shock nor its interaction with the female dummy is statistically significant, suggesting that tariff reductions do not meaningfully affect gender income disparity among the skilled. In contrast, both coefficients are statistically significant in the unskilled subsample: input tariff reductions significantly raise male workers' income, while the gains for female workers are substantially smaller, resulting in a widened gender income gap.

We next decompose income into hourly wages and working time. Columns (3) and (4) show that, for both skilled and unskilled workers, input tariff reductions increase gender wage inequality. In both subsamples, the interaction term between the tariff shock and the female dummy is positive and statistically significant, with comparable magnitudes,

indicating that female workers experience smaller wage gains than their male counterparts regardless of skill level.

The results for working time, reported in Columns (5) and (6), reveal an important difference across skill groups. Among skilled workers, input tariff reductions have no significant effect on relative working time. Among unskilled workers, however, tariff reductions significantly increase female workers' working time relative to that of male workers.

Finally, Columns (7) and (8) examine health outcomes. Consistent with the earlier patterns, the interaction term is insignificant for skilled workers but significantly negative for unskilled workers. This implies that input tariff reductions widen the gender health gap only among the unskilled.

Taken together, the results in Table 10 underscore the central role of education in shaping the gendered effects of trade liberalization. Although input tariff reductions tend to disadvantage female workers on average, the adverse impacts are concentrated overwhelmingly among the unskilled, while skilled workers appear largely insulated from these effects.

7 Conclusions

This paper studies the effects of input tariff liberalization on gender disparities, using China's accession to the WTO as a natural experiment. Descriptive evidence shows that, following WTO accession, female manufacturing workers experienced slower income growth, larger increases in working hours, and a higher probability of illness or injury relative to their male counterparts. To rationalize these patterns, we develop a simple yet flexible theoretical framework that highlights the role of gender differences in labor supply elasticities in shaping heterogeneous responses to trade shocks.

Guided by the model's predictions, we exploit regional variation in exposure to input tariff reductions and show that tariff liberalization raises incomes for all manufacturing workers. However, the income gains are significantly smaller for women, leading to a widening gender income gap. Further decomposition reveals that this divergence is not driven by a relative increase in male working time, but rather by a growing gender wage gap. These findings are consistent with female workers exhibiting more elastic labor supply responses to changes in production costs.

We also document a pronounced impact of input tariff liberalization on gender health inequality. Female manufacturing workers in prefectures more exposed to tariff reductions experience a significantly higher likelihood of illness or injury relative to male workers. A comprehensive set of robustness checks confirms the stability of these results. Moreover, we find that gender-specific labor demand shocks do not generate materially different effects, lending support to labor supply elasticities—as opposed to demand-side composition effects—as the primary mechanism underlying the observed gender disparities.

The adverse gender effects of input tariff liberalization are highly heterogeneous across education groups. The widening gender gaps in income and health are concentrated almost entirely among unskilled workers, while skilled workers are largely unaffected. This pattern underscores the protective role of education in mitigating the gender-specific costs of trade shocks. In light of the long-term trend toward narrowing gender gaps in educational attainment in China, our findings suggest that the trade-induced deterioration in gender equality may diminish over time.

This study contributes to several strands of the literature. First, it enriches the growing body of work on trade liberalization and gender inequality by identifying input tariff reductions as a previously underexplored channel through which trade policy affects gender outcomes. Second, by examining not only income and wages but also health outcomes, we

provide novel evidence on the health consequences of trade liberalization and their gendered nature. Finally, our analysis complements the literature on the distributional effects of China's WTO accession. While most existing studies emphasize changes in the skill premium, we show that WTO accession also had substantial and persistent effects on gender disparities.

References

Aguayo-Tellez, E., Airola, J., Juhn, C., Villegas-Sanchez, C. 2013. Did trade liberalization help women? the case of Mexico in the 1990s. Pp. 1–35 in: *New analyses in worker well-being*, vol. 38. United Kingdom: Emerald Group Publishing Limited. https://doi.org/10.1108/S0147-9121(2013)0000038001

Alesina, A., Ichino, A., Karabarbounis, L. 2011. Gender-based taxation and the division of family chores. *Am. Econ. J.: Econ. Policy* 3: 1–40. https://doi.org/10.1257/pol.3.2.1

Amiti, M., Davis, D.R. 2012. Trade, firms, and wages: theory and evidence. *Rev. Econom. Stud.* 79: 1–36. https://doi.org/10.1093/restud/rdr016

Amiti, M., Konings, J. 2007. Trade liberalization, intermediate inputs, and productivity: evidence from Indonesia. *Am. Econ. Rev.* 97: 1611–1638. https://doi.org/10.1257/aer.97.5.1611

Antràs, P., Fort, T.C., Tintelnot, F. 2017. The Margins of Global Sourcing: Theory and Evidence from US Firms. *Am. Econ. Rev.* 107: 2514–2564. https://doi.org/10.1257/aer.20141685

Autor, D., Dorn, D., Hanson, G. 2019. When work disappears: manufacturing decline and the falling marriage market value of young men. *Am. Econ. Rev.: Insights* 1: 161–178. https://doi.org/10.1257/aeri.20180010

Bai, X., Krishna, K., Ma, H. 2017. How you export matters: export mode, learning and productivity in China. *J. Int. Econ.* 104: 122–137. https://doi.org/10.1016/j.jinteco.2016.10.009

Bas, M., Strauss-Kahn, V. 2015. Input-trade liberalization, export prices and quality upgrading. *J. Int. Econ.* 95: 250–262. https://doi.org/10.1016/j.jinteco.2014.12.005

Berniell, I., Bietenbeck, J. 2020. The effect of working hours on health. *Econ. Hum. Biol.* 39: 100901. https://doi.org/10.1016/j.ehb.2020.100901

Blundell, R., Macurdy, T. 1999. Labor supply: a review of alternative approaches. Pp. 1559–1695 in: *Handbook of labor economics*, vol. 3. Elsevier. https://doi.org/10.1016/S1573-4463(99)03008-4

Bombardini, M., Li, B. 2020. Trade, pollution and mortality in China. *J. Int. Econ.* 125: 103321. https://doi.org/10.1016/j.jinteco.2020.103321

Brambilla, I., Lederman, D., Porto, G. 2012. Exports, export destinations, and skills. *Am. Econ. Rev.* 102: 3406–3438. https://doi.org/10.1257/aer.102.7.3406

Brandt, L., Van Biesebroeck, J., Wang, L., Zhang, Y. 2017. WTO Accession and Performance of Chinese Manufacturing Firms. *Am. Econ. Rev.* 107: 2784–2820. https://doi.org/10.1257/aer.20121266

Bustos, P. 2011. Trade Liberalization, Exports, and Technology Upgrading: Evidence on the Impact of MERCOSUR on Argentinian Firms. *Am. Econ. Rev.* 101: 304–340. https://doi.org/10.1257/aer.101.1.304

Dudek, M. 2025. Seemingly baseless discrimination. *Games Econom. Behav.* 149: 43–55. https://doi.org/10.1016/j.geb.2024.11.003

Duflo, E. 2012. Women empowerment and economic development. *J. Econ. Lit.* 50: 1051–1079. https://doi.org/10.1257/jel.50.4.1051

Eaton, J., Kortum, S. 2002. Technology, Geography, and Trade. *Econometrica* 70: 1741–1779. https://doi.org/10.1111/1468-0262.00352

Erten, B., Leight, J. 2021. Exporting out of agriculture: the impact of WTO accession on structural transformation in China. *Rev. Econ. Stat.* 103: 364–380. https://doi.org/10.1162/rest_a_00852

Fan, H., Li, Y.A., Yeaple, S.R. 2015. Trade liberalization, quality, and export prices. *Rev. Econ. Stat.* 97: 1033–1051. https://doi.org/10.1162/REST_a_00524

Fan, H., Li, Y.A., Yeaple, S.R. 2018. On the relationship between quality and productivity: evidence from China's accession to the WTO. *J. Int. Econ.* 110: 28–49. https://doi.org/10.1016/j.jinteco.2017.10.001

Fan, H., Lin, F., Lin, S. 2020. The hidden cost of trade liberalization: input tariff shocks and worker health in China. *J. Int. Econ.* 126: 103349. https://doi.org/10.1016/j.jinteco.2020.103349

Feenstra, R.C., Hanson, G.H. 1999. The impact of outsourcing and high-technology capital on wages: estimates for the United States, 1979–1990*. *Q. J. Econ.* 114: 907–940. https://doi.org/10.1162/003355399556179

Goldberg, P.K., Khandelwal, A.K., Pavcnik, N., Topalova, P. 2010. Imported intermediate inputs and domestic product growth: evidence from India*. *Q. J. Econ.* 125: 1727–1767. https://doi.org/10.1162/qjec.2010.125.4.1727

Goldberg, P.K., Pavcnik, N. 2005. Trade, wages, and the political economy of trade protection: evidence from the Colombian trade reforms. *J. Int. Econ.* 66: 75–105. https://doi.org/10.1016/j.jinteco.2004.04.005

Goldberg, P.K., Pavcnik, N. 2007. Distributional effects of globalization in developing countries. *J. Econ. Lit.* 45: 39–82. https://doi.org/10.1257/jel.45.1.39

Goussé, M., Jacquemet, N., Robin, J.-M. 2017. Marriage, labor supply, and home production. *Econometrica* 85: 1873–1919. https://doi.org/10.3982/ECTA11221

Guo, R., Lin, L., Yi, J., Zhang, J. 2020. The cross-spousal effect of education on health. *J. Dev. Econ.* 146: 102493. https://doi.org/10.1016/j.jdeveco.2020.102493

Halpern, L., Koren, M., Szeidl, A. 2015. Imported inputs and productivity. *Am. Econ. Rev.* 105: 3660–3703. https://doi.org/10.1257/aer.20150443

Hamermesh, D.S., Kawaguchi, D., Lee, J. 2017. Does labor legislation benefit workers? Well-being after an hours reduction. *J. Jpn. Int. Econ.* 44: 1–12. https://doi.org/10.1016/j.jjie.2017.02.003

Han, J., Liu, R., Ural Marchand, B., Zhang, J. 2016. Market structure, imperfect tariff pass-through, and household welfare in Urban China. *J. Int. Econ.* 100: 220–232. https://doi.org/10.1016/j.jinteco.2016.02.006

Han, J., Liu, R., Zhang, J. 2012. Globalization and wage inequality: evidence from urban China. *J. Int. Econ.* 87: 288–297. https://doi.org/10.1016/j.jinteco.2011.12.006

Handley, K., Limão, N. 2017. Policy uncertainty, trade, and welfare: theory and evidence

for China and the United States. *Am. Econ. Rev.* 107: 2731–2783. https://doi.org/10.1257/aer.20141419

Helpman, E., Itskhoki, O., Redding, S. 2010. Inequality and unemployment in a global economy. *Econometrica* 78: 1239–1283. https://doi.org/10.3982/ECTA8640

Huang, C., Li, H., Liu, P.W., Zhang, J. 2009. Why does spousal education matter for earnings? Assortative mating and cross-productivity. *J. Labor Econ.* 27: 633–652. https://doi.org/10.1086/644746

Hummels, D., Munch, J., Xiang, C. 2016, July. No pain, No gain: work demand, work effort, and worker health. Working Paper, National Bureau of Economic Research. https://doi.org/10.3386/w22365

Hummels, D., Munch, J.R., Xiang, C. 2025. No pain, No gain: work demand, work effort, and worker health. *Rev. Econ. Stat.* 107: 898–913. https://doi.org/10.1162/rest_a_01335

Juhn, C., Ujhelyi, G., Villegas-Sanchez, C. 2013. Trade liberalization and gender inequality. *Am. Econ. Rev.* 103: 269–273. https://doi.org/10.1257/aer.103.3.269

Juhn, C., Ujhelyi, G., Villegas-Sanchez, C. 2014. Men, women, and machines: how trade impacts gender inequality. *J. Dev. Econ.* 106: 179–193. https://doi.org/10.1016/j.jdeveco.2013.09.009

Kasahara, H., Rodrigue, J. 2008. Does the use of imported intermediates increase productivity? Plant-level evidence. *J. Dev. Econ.* 87: 106–118. https://doi.org/10.1016/j.jdeveco.2007.12.008

Keane, M.P. 2011. Labor supply and taxes: a survey. *J. Econ. Lit.* 49: 961–1075. https://doi.org/10.1257/jel.49.4.961

Keane, M.P. 2022. Recent research on labor supply: implications for tax and transfer policy. *Labour Econ.* 77: 102026. https://doi.org/10.1016/j.labeco.2021.102026

Killingsworth, M.R., Heckman, J.J. 1986. Chapter 2 Female labor supply: A survey. Pp. 103–204 in: *Handbook of Labor Economics*, vol. 1. Elsevier. https://doi.org/10.1016/S1573-4463(86)01005-2

Kovak, B.K. 2013. Regional effects of trade reform: what is the correct measure of liberalization? *Am. Econ. Rev.* 103: 1960–1976. https://doi.org/10.1257/aer.103.5.1960

Li, H. 2023. *Human capital in China 2023*. Beijing, China: Center for Human Capital and Labor Market Research, Central University of Finance and Economics.

Lucas, R.E. 1978. On the size distribution of business firms. *Bell J. Econ.* 9: 508–523. https://doi.org/10.2307/3003596

Lundberg, S. 1985. The added worker effect. *J. Labor Econ.* 3: 11–37. https://doi.org/10.1086/298069

McClelland, R., Mok, S. 2012, October 1. A review of recent research on labor supply elasticities. Working Paper, Washington, DC: Congressional Budget Office. https://hdl.handle.net/1813/77792

McManus, T.C., Schaur, G. 2016. The effects of import competition on worker health. *J. Int.*

Econ. 102: 160–172. https://doi.org/10.1016/j.jinteco.2016.06.003

Oster, E. 2012. Routes of Infection: Exports and HIV Incidence in Sub-Saharan Africa. *J. Eur. Econ. Assoc.* 10: 1025–1058. https://doi.org/10.1111/j.1542-4774.2012.01075.x

Ozler, S. 2000. Export orientation and female share of employment: evidence from Turkey. *World Dev.* 28: 1239–1248. https://doi.org/10.1016/S0305-750X(00)00034-6

Pavcnik, N. 2017, September. The impact of trade on inequality in developing countries. Working Paper, National Bureau of Economic Research. https://doi.org/10.3386/w23878

Pierce, J.R., Schott, P.K. 2016. The surprisingly swift decline of US manufacturing employment. *Am. Econ. Rev.* 106: 1632–1662. https://doi.org/10.1257/aer.20131578

Rendall, M. 2010. Brain versus brawn: the realization of women's comparative advantage. *SSRN Electron. J.* 0–53. https://doi.org/10.2139/ssrn.1635251

Rod, N.H., Kjeldgård, L., Åkerstedt, T., Ferrie, J.E., Salo, P., Vahtera, J., Alexanderson, K. 2017. Sleep Apnea, Disability Pensions, and Cause-Specific Mortality: A Swedish Nationwide Register Linkage Study. *Am. J. Epidemiol.* 186: 709–718. https://doi.org/10.1093/aje/kwx138

Setzler, B., Tintelnot, F. 2021. The effects of foreign multinationals on workers and firms in the United States*. *Q. J. Econ.* 136: 1943–1991. https://doi.org/10.1093/qje/qjab015

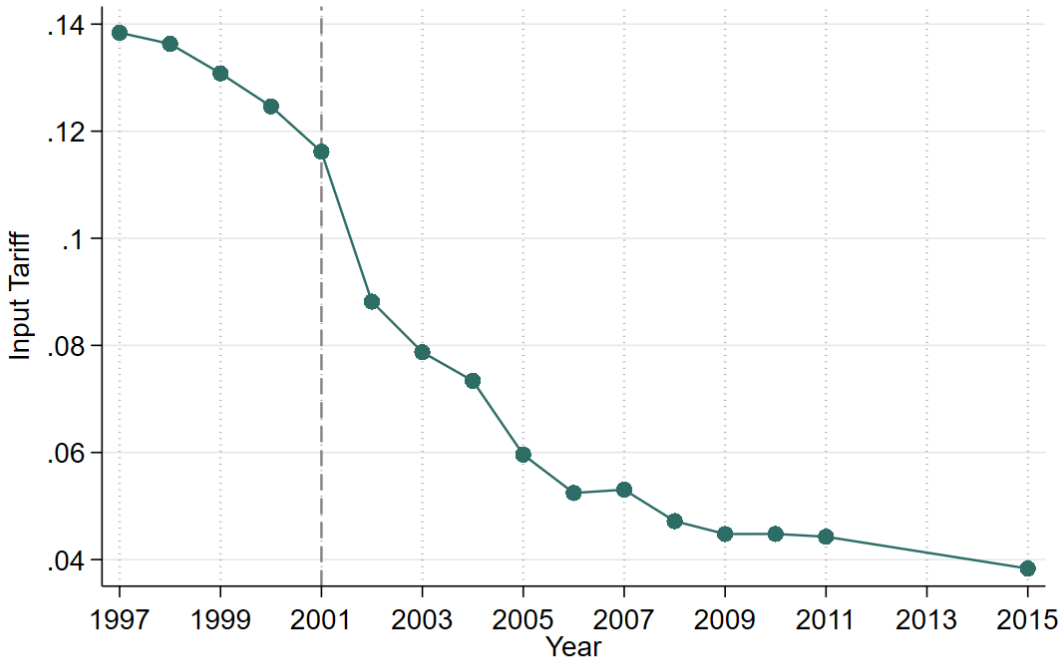
Topalova, P. 2010. Factor immobility and regional impacts of trade liberalization: evidence on poverty from India. *Am. Econ. J.: Appl. Econ.* 2: 1–41. https://doi.org/10.1257/app.2.4.1

Verhoogen, E.A. 2008. Trade, Quality Upgrading, and Wage Inequality in the Mexican Manufacturing Sector. *Q. J. Econ.* 123: 489–530. https://doi.org/10.1162/qjec.2008.123.2.489

Wang, F., Kis-Katos, K., Zhou, M. 2022. Import competition and the gender employment gap in China. *J. Hum. Resour.* 1220. https://doi.org/10.3368/jhr.1220-11399R3

Weinberg, B.A. 2000. Computer use and the demand for female workers. *ILR Rev.* 53: 290–308. https://doi.org/10.1177/001979390005300206

Figure 1. Input Tariff Trend in China



Source: WITS, 2002 IO table and ASIF

Figure 2. Chinese Prefectures' Exposures to the Input Tariff Shock

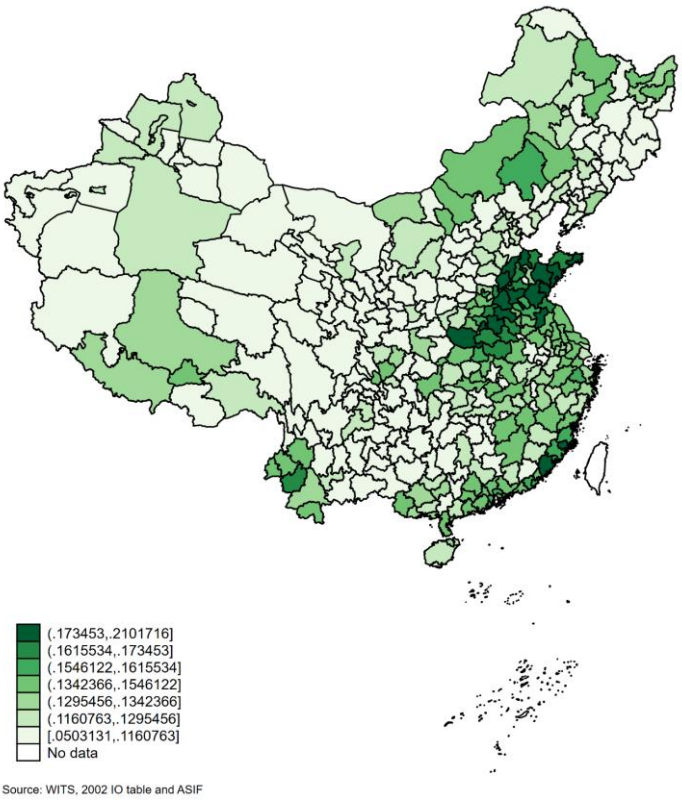
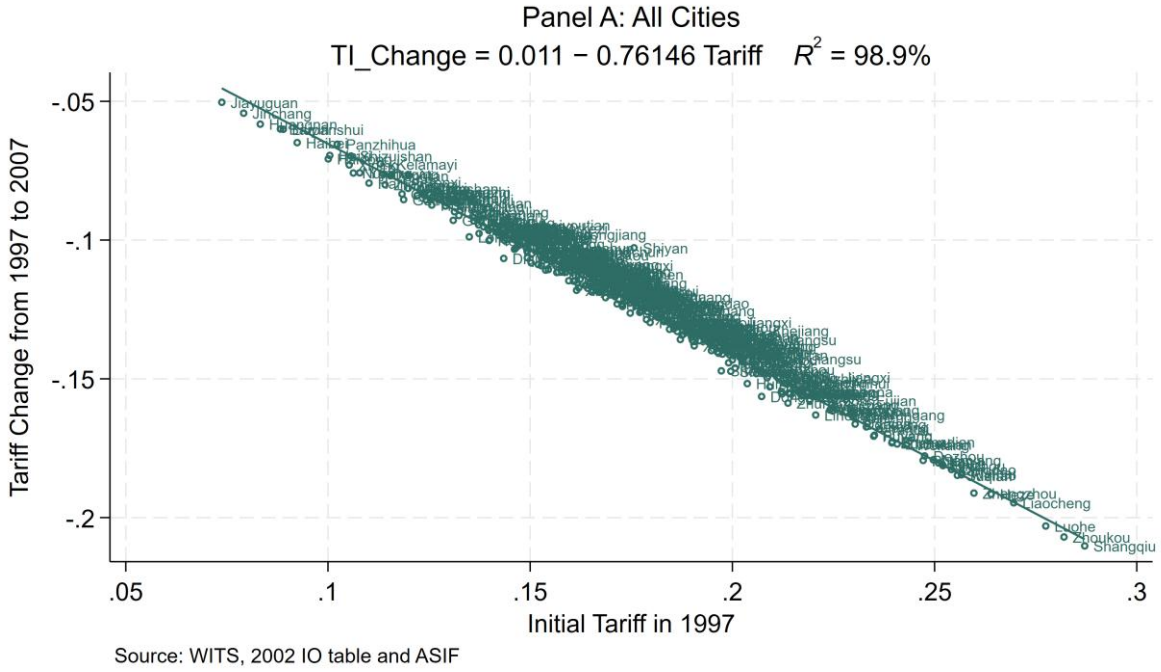


Figure 3. Initial Tariff Level and Chang in Tariff



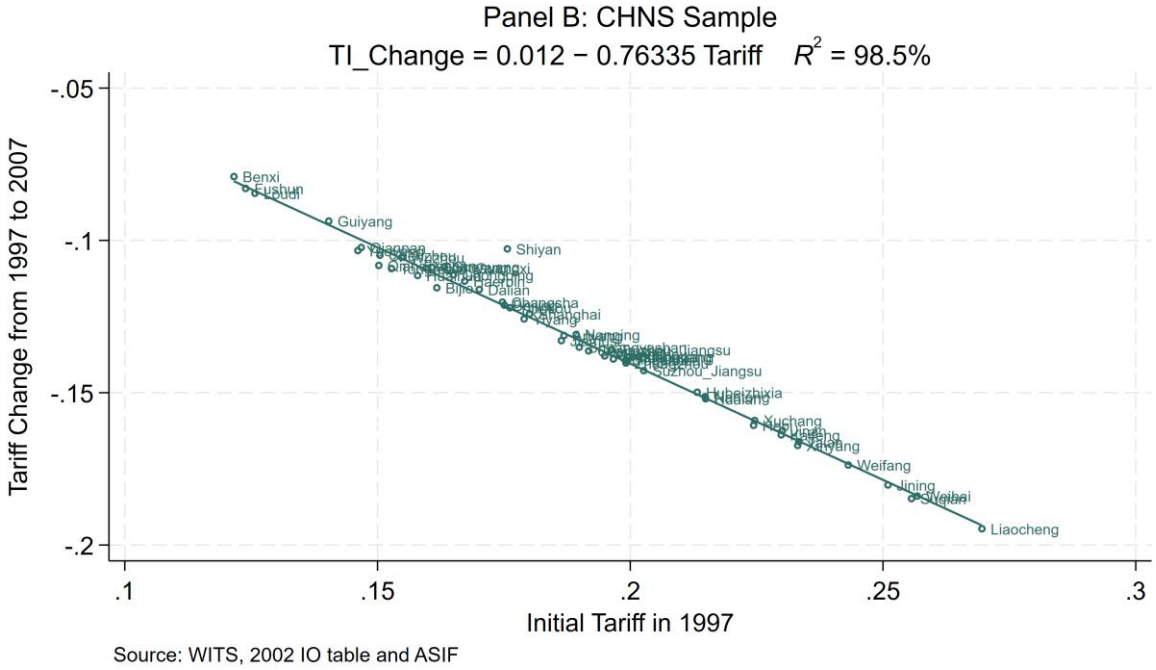


Figure 4. Estimated DID Coefficients

Table 1. Before and After Mean Comparisons

Variables	Before WTO			After WTO			Change
	Male	Female	Difference	Male	Female	Difference	DID
(1) Log(Yearly Income (CHNS))	8.991	8.823	0.168	9.881	9.576	0.305	0.137
(2) Log(Yearly Income (UHS))	8.818	8.571	0.246	9.421	9.118	0.303	0.057
(3) Log(Hourly Wage)	1.360	1.209	0.151	2.261	1.922	0.339	0.248
(4) Log(Working time)	3.808	3.781	0.028	3.857	3.890	-0.033	-0.061
(5) Illness or Injury	0.056	0.050	0.006	0.098	0.123	-0.024	-0.03

Table 2. The Effects on Income, Wage, and Working Time

	(1) Income_CHNS	(2) Income_UHS	(3) Hourly Wage	(4) Working Time
Tariff	-2.776**	-2.666**	-2.580**	-0.584
	(1.266)	(1.113)	(1.036)	(0.389)
Tariff*Female	0.852***	1.035***	1.224***	-0.408***
	(0.318)	(0.146)	(0.298)	(0.108)
Female	-0.343***	-0.334***	-0.382***	0.035**
	(0.042)	(0.014)	(0.035)	(0.015)
Eduyear	0.014***	0.041***	0.008**	-0.005***
	(0.004)	(0.001)	(0.003)	(0.001)
Age	0.028***	0.086***	0.025***	-0.007***
	(0.008)	(0.003)	(0.006)	(0.003)
Agesquare	-0.000***	-0.001***	-0.000***	0.000*
	(0.000)	(0.000)	(0.000)	(0.000)
Urban	0.050		0.093**	-0.003
	(0.051)		(0.045)	(0.028)
Married	0.079**		0.042*	0.032***
	(0.032)		(0.025)	(0.012)
Minority	-0.016		0.008	0.006
	(0.040)		(0.042)	(0.018)
Tariff_Output	2.519**	6.078***	1.713	1.477***
	(1.226)	(1.169)	(1.109)	(0.339)
Exportpolicyuncertainty	0.189***	-0.025	0.190***	-0.021
	(0.065)	(0.021)	(0.060)	(0.025)
Tradelicensepolicy	0.303	-0.013	0.102	0.116
	(0.293)	(0.020)	(0.285)	(0.111)
Quotapolicy	-0.152***	-0.005	-0.164***	0.006
	(0.056)	(0.011)	(0.052)	(0.022)
Constant	8.365***	6.072***	0.760**	3.844***
	(0.344)	(0.151)	(0.352)	(0.131)
Observations	5,909	81,268	5,844	6,761
R-squared	0.465	0.445	0.557	0.160
Occupation Type	Y	Y	Y	Y
Employer Ownership	Y	Y	Y	Y
Year FE	Y	Y	Y	Y
Prefecture FE	Y	Y	Y	Y

Notes: All regressions include output tariff shock, a set of individual characteristics, other trade policy shocks, a constant, prefecture fixed effects, and year fixed effects. Column (1) shows the effect on yearly income with the CHNS sample, and Column (2) shows the result on yearly income with the UHS sample. Column (3) shows the effect on hourly wage, and Column (4) shows the result on working time. Standard errors clustered at the prefecture-year level are in parentheses. ***, **, and * indicate significance levels of 1%, 5%, and 10%, respectively.

Table 3. The Effect on Health

	(1)	(2)	(3)	(4)	(5)	(6)
	Linear	Logit	Linear	logit	Linear	Logit
Tariff	-0.365	-6.941	0.721	10.033	0.795	10.823
	(0.385)	(6.454)	(0.653)	(8.712)	(0.665)	(8.870)
Tariff*Female	-0.359***	-5.683***	-0.404**	-6.247***	-0.539**	-7.908***
	(0.113)	(1.745)	(0.167)	(2.393)	(0.209)	(2.673)
Female	0.057***	0.770***	0.033	0.478*	0.369	4.614*
	(0.018)	(0.213)	(0.024)	(0.273)	(0.243)	(2.616)
Disease History	0.116***	0.958***	0.133***	1.070***	0.132***	1.062***
	(0.022)	(0.147)	(0.028)	(0.179)	(0.028)	(0.179)
Smoking	0.003	0.057	-0.003	-0.063	-0.004	-0.071
	(0.009)	(0.125)	(0.016)	(0.186)	(0.016)	(0.186)
Abnormal BMI	0.003	0.031	-0.000	-0.007	-0.001	-0.009
	(0.007)	(0.086)	(0.011)	(0.120)	(0.011)	(0.120)
Health Insurance	0.016*	0.248*	0.022	0.320	0.023	0.323*
	(0.010)	(0.135)	(0.015)	(0.196)	(0.015)	(0.196)
Log(Air Quality Index)	0.043	1.036	0.092*	1.148	0.090	1.141
	(0.044)	(0.982)	(0.054)	(1.258)	(0.055)	(1.264)
Yearly Income			-0.035***	-0.365***	-0.028**	-0.277**
			(0.010)	(0.096)	(0.011)	(0.120)
Labor Burden			0.046**	0.503**	0.067**	0.793**
			(0.021)	(0.249)	(0.031)	(0.365)
Female * Income					-0.016	-0.177
					(0.016)	(0.157)
Female * Labor Burden					-0.042	-0.574
					(0.045)	(0.521)
Constant	-0.006	-5.059	-0.024	-2.854	-0.176	-4.823
	(0.197)	(4.786)	(0.267)	(6.294)	(0.293)	(6.527)
Observations	7,207	7,137	3,281	3,160	3,281	3,160
R-squared	0.051		0.064		0.065	
Individual Controls	Y	Y	Y	Y	Y	Y
Other Trade Policy	Y	Y	Y	Y	Y	Y
Occupation Type	Y	Y	Y	Y	Y	Y
Employer Ownership	Y	Y	Y	Y	Y	Y
Year FE	Y	Y	Y	Y	Y	Y
Prefecture FE	Y	Y	Y	Y	Y	Y

Notes: All regressions include a constant, full set of control variables in the benchmark regression, occupation type dummies, employer ownership dummies, prefecture fixed effects, and year fixed effects. To ensure the effects are not driven by the individuals' initial health conditions, we additionally control four health-related variables. The outcome is namely the dummy for all types of illness or injury mentioned in the survey. Columns (3) – (4) further control for annual income and labor burden (defined as the sum of time spent on paid employment and domestic chores). Columns (5) – (6) include these two variables as well as their interaction terms with the female dummy. Standard errors clustered at the prefecture-year level are in parentheses. ***, **, and * indicate significance levels of 1%, 5%, and 10%, respectively.

Table 4. Controlling for Additional Interaction Effects

	(1) Income	(2) Hourly Wage	(3) Working Time	(4) Illness or Injury
Tariff	-5.028**	-2.442	-0.623	-2.126***
	(2.336)	(1.810)	(0.855)	(0.805)
Tariff*Female	0.704**	1.137***	-0.361***	-0.315***
	(0.331)	(0.302)	(0.110)	(0.115)
Tariff*Age	0.234*	0.086	-0.014	0.063
	(0.124)	(0.093)	(0.043)	(0.040)
Tariff*Age Square	-0.003**	-0.001	0.000	-0.001
	(0.002)	(0.001)	(0.001)	(0.001)
Tariff*Education	-0.161***	-0.162***	0.005	0.048**
	(0.053)	(0.047)	(0.020)	(0.020)
Observations	5,909	5,844	6,761	7,207
R-squared	0.467	0.558	0.161	0.053
Individual Controls	Y	Y	Y	Y
Other Trade Policy	Y	Y	Y	Y
Occupation Type	Y	Y	Y	Y
Employer Ownership	Y	Y	Y	Y
Year FE	Y	Y	Y	Y
Prefecture FE	Y	Y	Y	Y

Notes: All regressions include a constant, prefecture fixed effects, year fixed effects, a set of individual characteristics, and trade policy shocks. To isolate the effects from the interaction between tariff change and individual characteristics, we additionally control three interaction terms: $Tariff * Age$, $Tariff * Age\ Square$ and $Tariff * Education$. Standard errors clustered at the prefecture-year level are in parentheses. ***, **, and * indicate significance levels of 1%, 5%, and 10%, respectively.

Table 5. Controlling for Spouse Characteristics in Married Subsamples

	(1)	(2)	(3)	(4)	(5)
			CHNS		UHS
	Income	Hourly Wage	Working Time	Illness or Injury	Income
Tariff	-0.982	-2.422*	-0.350	-0.534	-1.062
	(1.596)	(1.340)	(0.481)	(0.473)	(1.386)
Tariff * Female	0.603*	1.173***	-0.515***	-0.411***	0.446*
	(0.357)	(0.326)	(0.129)	(0.130)	(0.231)
female	-0.368***	-0.399***	0.042**	0.057***	-0.347***
	(0.046)	(0.039)	(0.017)	(0.021)	(0.026)
Age^{spouse}	0.002	0.001	0.000	0.003**	-0.001***
	(0.003)	(0.003)	(0.001)	(0.001)	(0.000)
$Education^{spouse}$	0.003	0.000	-0.003	-0.001	0.012***
	(0.004)	(0.003)	(0.002)	(0.001)	(0.002)
$Employment^{spouse}$	0.031	0.012	-0.004	-0.023**	0.013
	(0.025)	(0.019)	(0.010)	(0.010)	(0.010)
Constant	8.801***	0.779*	3.937***	0.024	6.485***
	(0.395)	(0.399)	(0.170)	(0.226)	(0.193)
Observations	4,441	4,398	5,072	5,369	27,444
R-squared	0.482	0.556	0.179	0.049	0.458
Occupation Type	Y	Y	Y	Y	Y
Employer Ownership	Y	Y	Y	Y	Y
Year FE	Y	Y	Y	Y	Y
Prefecture FE	Y	Y	Y	Y	Y

Notes: This table reports the results from subsamples that include only married individuals. Columns (1) – (4) use the CHNS subsample, and Column (5) uses the UHS subsample. All regressions include a constant, prefecture fixed effects, year fixed effects, a set of individual characteristics, and trade policy shocks. In addition, all regressions also control for three spouse characteristics: age, education, and employment status. Standard errors clustered at the prefecture-year level are in parentheses. ***, **, and * indicate significance levels of 1%, 5%, and 10%, respectively.

Table 6. Excluding Outlier and Controlling for Prefecture Linear Trend

	(1) Income	(2) Hourly Wage	(3) Working Time	(4) Illness or Injury
		Panel A: Exclude Outlier		
Tariff	-2.928**	-2.813***	-0.632	-0.351
	(1.262)	(1.035)	(0.396)	(0.384)
Tariff*Female	0.854***	1.211***	-0.414***	-0.374***
	(0.321)	(0.301)	(0.109)	(0.113)
Observations	5,713	5,646	6,540	6,976
R-squared	0.465	0.560	0.159	0.053
		Panel B: Prefecture Linear Trend		
Tariff	-0.811	-0.331	0.222	-2.131*
	(3.622)	(2.714)	(1.196)	(1.131)
Tariff*Female	0.736**	1.129***	-0.382***	-0.336***
	(0.327)	(0.310)	(0.118)	(0.123)
Observations	5,909	5,844	6,761	7,207
R-squared	0.479	0.566	0.170	0.060
Individual Controls	Y	Y	Y	Y
Other Trade Policy	Y	Y	Y	Y
Occupation Type	Y	Y	Y	Y
Employer Ownership	Y	Y	Y	Y
Year FE	Y	Y	Y	Y
Prefecture FE	Y	Y	Y	Y

Notes: All regressions include a constant, prefecture fixed effects, year fixed effects, a set of individual characteristics, and trade policy shocks. All regressions in Panel A exclude the outlier city, Shiyan, from the sample. The regressions in Panel B conduct an analysis by additionally controlling the prefecture-wave fixed effect. Outcomes are namely the yearly income, the hourly wage, the weekly working time, and the dummy for illness or injury. Standard errors clustered at the prefecture-year level are in parentheses in Panel A and at the prefecture level are in parentheses in Panel B. ***, **, and * indicate significance levels of 1%, 5%, and 10%, respectively.

Table 7. IV Regression Results

	(1) Income_CHNS	(2) Income_UHS	(3) Hourly Wage	(4) Working Time	(5) Illness or Injury
Tariff	-2.484*	-2.121*	-2.410**	-0.894*	-0.260
	(1.356)	(1.285)	(1.082)	(0.455)	(0.425)
Tariff*Female	1.265***	0.878***	1.623***	-0.310**	-0.299**
	(0.378)	(0.207)	(0.345)	(0.123)	(0.125)
female	-0.385***	-0.323***	-0.423***	0.025	0.051***
	(0.048)	(0.020)	(0.041)	(0.016)	(0.019)
Observations	5,909	60,117	5,844	6,761	7,207
R-squared	0.409	0.321	0.505	0.076	0.029
Individual Controls	Y	Y	Y	Y	Y
Other Trade Policy	Y	Y	Y	Y	Y
Occupation Type	Y	Y	Y	Y	Y
Employer Ownership	Y	Y	Y	Y	Y
Year FE	Y	Y	Y	Y	Y
Prefecture FE	Y	Y	Y	Y	Y
Kleibergen-Paap rk Wald F statistic	49.34	111.1	48.88	48.01	49.32

Notes: This table reports the IV regression results. All regressions include output tariff shock, a set of individual characteristics, other trade policy shocks, a constant, prefecture fixed effects, and year fixed effects. Column (1) shows the effect on yearly income with the CHNS sample, and Column (2) shows the result on yearly income with the UHS sample. Column (3) shows the effect on hourly wage, and Column (4) shows the result on working time. Standard errors clustered at the prefecture-year level are in parentheses. ***, **, and * indicate significance levels of 1%, 5%, and 10%, respectively.

Table 8. Placebo Tests

	(1) Income	(2) Hourly Wage	(3) Working Time	(4) Illness or Injury
		Panel A: Agriculture		
Tariff	4.353**	2.789	-0.237	-0.970***
	(2.001)	(15.159)	(2.758)	(0.363)
Tariff*Female	-0.483	-2.694	0.082	0.038
	(0.442)	(1.648)	(0.343)	(0.065)
Observations	12,649	5,627	7,343	14,878
R-squared	0.267	0.317	0.190	0.047
		Panel B: Service		
Tariff	-5.422***	-5.238***	0.825	-0.517
	(1.609)	(1.371)	(0.755)	(0.687)
Tariff*Female	-1.169**	-1.279**	-0.048	-0.008
	(0.542)	(0.506)	(0.236)	(0.122)
Observations	2,688	2,669	4,088	4,736
R-squared	0.406	0.499	0.103	0.057
Individual Controls	Y	Y	Y	Y
Other Trade Policy	Y	Y	Y	Y
Occupation Type	Y	Y	Y	Y
Employer Ownership	Y	Y	Y	Y
Year FE	Y	Y	Y	Y
Prefecture FE	Y	Y	Y	Y

Notes: All regressions include a constant, a full set of control variables, occupation type dummies, employer ownership dummies, prefecture fixed effects and year fixed effects. Panel A is for the agriculture sector, and Panel B is for the service sector. Outcomes are the yearly income, the hourly wage, the weekly working time, and the dummy for illness or injury from Columns (1) to (4) for both panels. Standard errors clustered at the prefecture-year level are in parentheses. ***, **, and * indicate significance levels of 1%, 5%, and 10%, respectively.

Table 9. Gender-Specific Tariff Shock

	(1) Income	(2) Income	(3) Hourly Wage	(4) Hourly Wage	(5) Work Time	(6) Work Time	(7) Illness or Injury	(8) Illness or Injury
$Tariff^{Female}$	-3.850*		-3.613**		-1.353**		-0.289	
	(2.009)		(1.669)		(0.608)		(0.628)	
$Tariff^{Female}$ * Female	1.651**		2.352***		-0.846***		-0.763***	
	(0.657)		(0.627)		(0.231)		(0.233)	
$Tariff^{Male}$		-5.891**		-5.532**		-0.073		-0.985
		(2.708)		(2.210)		(0.885)		(0.819)
$Tariff^{Male}$ * Female		1.603***		2.333***		-0.728***		-0.640***
		(0.593)		(0.547)		(0.197)		(0.209)
female	-0.332***	-0.345***	-0.365***	-0.388***	0.032**	0.035**	0.055***	0.057***
	(0.040)	(0.043)	(0.033)	(0.036)	(0.014)	(0.015)	(0.017)	(0.018)
Constant	8.246***	8.482***	0.657*	0.866**	3.815***	3.816***	-0.051	0.025
	(0.333)	(0.361)	(0.338)	(0.371)	(0.125)	(0.143)	(0.189)	(0.204)
Observations	5,909	5,909	5,844	5,844	6,761	6,761	7,207	7,207
R-squared	0.465	0.465	0.556	0.557	0.160	0.159	0.051	0.051
Occupation Type	Y	Y	Y	Y	Y	Y	Y	Y
Employer Ownership	Y	Y	Y	Y	Y	Y	Y	Y
Year FE	Y	Y	Y	Y	Y	Y	Y	Y
Prefecture FE	Y	Y	Y	Y	Y	Y	Y	Y

Notes: All regressions include a constant, prefecture fixed effects, year fixed effects, a set of individual characteristics, and trade policy shocks. Columns (1), (3), (5) and (7) employ the female-dominated tariff shock, and Columns (2), (4), (6) and (8) employ the male-dominated tariff shock. Standard errors clustered at the prefecture-year level are in parentheses. ***, **, and * indicate significance levels of 1%, 5%, and 10%, respectively.

Table 10. The Role of Education: Skilled vs. Unskilled

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Income		Hourly Wage		Working Time		Illness or Injury	
	Skilled	Unskilled	Skilled	Unskilled	Skilled	Unskilled	Skilled	Unskilled
Tariff	-1.422	-3.017**	-0.006	-3.074***	-1.341***	-0.345	-0.486	-0.350
	(1.715)	(1.405)	(1.409)	(1.093)	(0.505)	(0.516)	(0.635)	(0.474)
Tariff*Female	0.511	1.014***	1.115**	1.266***	-0.203	-0.444***	-0.262	-0.382***
	(0.455)	(0.375)	(0.476)	(0.330)	(0.198)	(0.134)	(0.180)	(0.132)
Observations	1,910	3,999	1,886	3,958	2,089	4,672	2,172	5,035
R-squared	0.511	0.448	0.609	0.534	0.167	0.159	0.063	0.057
Individual Controls	Y	Y	Y	Y	Y	Y	Y	Y
Other Trade Policy	Y	Y	Y	Y	Y	Y	Y	Y
Occupation Type	Y	Y	Y	Y	Y	Y	Y	Y
Employer Ownership	Y	Y	Y	Y	Y	Y	Y	Y
Year FE	Y	Y	Y	Y	Y	Y	Y	Y
Prefecture FE	Y	Y	Y	Y	Y	Y	Y	Y

Notes: All regressions include a constant, prefecture fixed effects, year fixed effects, a set of individual characteristics, and trade policy shocks. Columns (1), (3), (5) and (7) are for individuals with a high school education level or above, and Columns (2), (4), (6) and (8) are for the otherwise. Standard errors clustered at the prefecture-year level are in parentheses. ***, **, and * indicate significance levels of 1%, 5%, and 10%, respectively.

Online Appendix (Not for Publication)

Data Appendix

Part A. The CHNS Sample

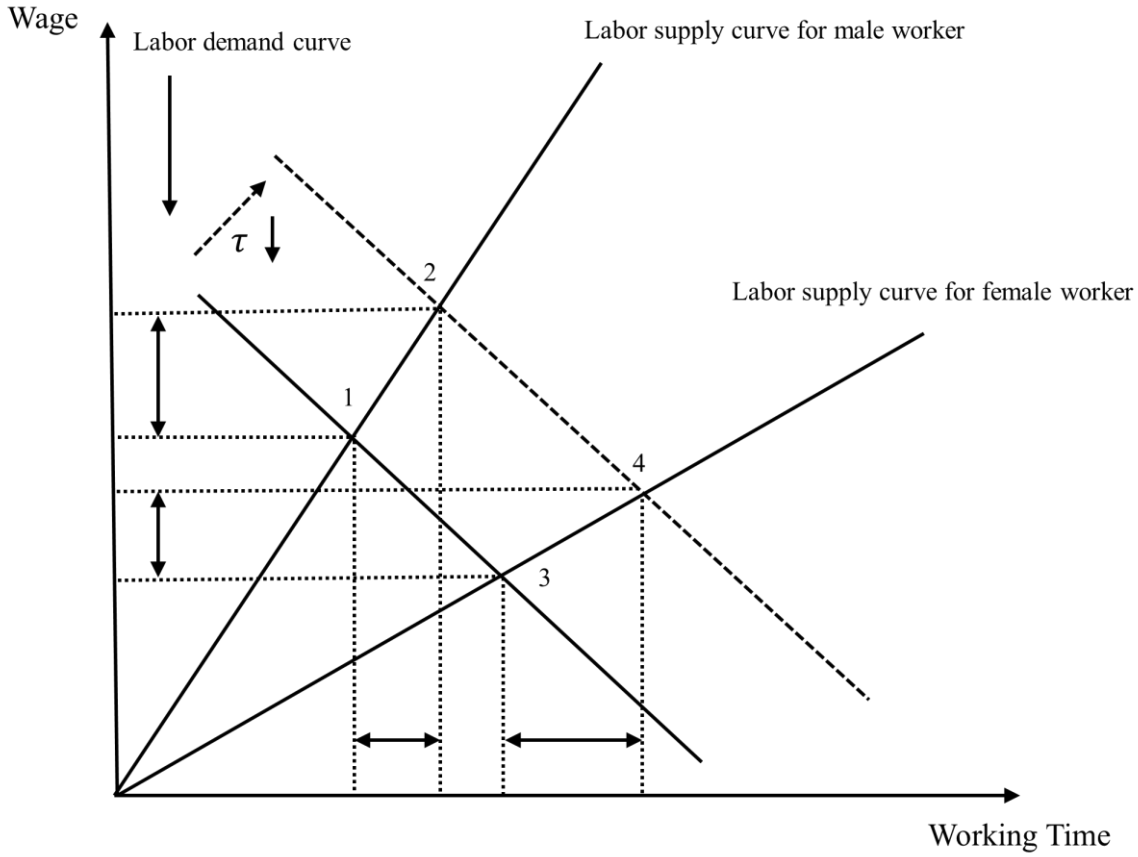
Panel A of Appendix Table 1 provides a list of occupations in the CHNS. Among the 13 occupations, we use individuals whose occupations are coded as 6 or 7 as manufacturing workers in our main analysis. We exclude individuals under 16 or above 55 (60) for females (males), and individuals whose weekly working hours are less than 10. In our robustness checks, we consider individuals whose occupations are coded as 5 as workers in the agriculture sector and those coded as 11 as workers in the service sector.

Part B. The UHS Sample

Panel B of Appendix Table 1 provides a list of occupations in the UHS. Among the 16-20 industries available[15], we restrict our sample to the workers in the manufacturing sector (Code = 3). We also exclude individuals under 16 or above 55 (60) for females (males) and those who did not report any job information from the sample.

[15] For the waves until 2005, there are 16 industries available; For the waves since 2006, there are 20 industries. But the code for manufacturing sector is always 3 in our sample waves.

Appendix Figure 1 Gender Difference in Labor Supply



Appendix Table 1 Industry Codes

Panel A: CHNS Code

Code	Description of occupation
1	Senior professional technical workers (professor, doctor, lawyer, etc.)
2	General professional technical workers (teacher, nurse, editor, etc.)
3	Officials and administrators (government workers)
4	Office workers (secretary, clerk)
5	Farmers, fishers, hunters
6	Experienced factory workers
7	Less experienced factory workers
8	Military officers and high-ranked police officers
9	Soldiers and low-ranked police officers
10	Drivers
11	Service workers (barber, cook, shop assistant, etc.)
12	Athletes, actors, performers
13	Others

Panel B: UHS Code

Until 2005		Since 2006	
Code	Industry Type	Code	Industry Type
1	Agriculture, forestry, animal husbandry and fishery	1	Agriculture, forestry, animal husbandry and fishery
2	Extractive industry	2	Extractive industry
3	Manufacturing industry	3	Manufacturing industry
4	Electricity, gas and water production and supply industry	4	Electricity, gas and water production and supply industry
5	Construction industry	5	Construction industry
6	Geological survey industry, water management industry	6	Transportation, storage and post and telecommunications
7	Transportation, storage and post and telecommunications	7	Information transmission, computer services and software industry
8	Wholesale and retail, catering industry	8	Wholesale and retail industry
9	Financial Industry	9	Accommodation and catering industry
10	Real Estate industry	10	Financial Industry
11	Social Services	11	Real Estate industry
12	Health, sports and social welfare industry	12	Rental and business services
13	Education, culture and arts and radio, film and television industry	13	Scientific research, integrated technical services and the geological survey industry
14	Scientific research and integrated technical services	14	Water, environment and public facilities management industry
15	Government institutions and social organizations	15	Residential services and other services
16	Others	16	Education

	17	Health, social security and social welfare industry
	18	Culture, Sports and Entertainment industry
	19	Government institutions and social organizations
	20	International organizations

Appendix Table 2 Variable Definitions and Data Sources

Variables	Definitions	Data Sources
Log(Income(CHNS))	Annual wage income plus bonus winsorized 1% in each tail and adjusted by regionally different CPI to 2015 price level in RMB in log	CHNS
Log(Income(UHS))	Annual working income winsorized 1% in each tail and adjusted by national CPI to 2008 price level in RMB in log	UHS
Log(Working Time)	Log weekly hours worked	CHNS
Log(Hourly Wage)	Average hourly wage winsorized 1% in each tail and adjusted by regionally different CPI to 2015 price level in log	CHNS
Illness or Injury	Dummy for illness or injury in the last four weeks	CHNS
Education, Age, Female, Married, Urban, Minority	Years of education, age, female, married, urban and minority dummy	CHNS, UHS
Occupation Type, Employer Ownership	Categories for occupation types and types of employer ownership	CHNS, UHS
Disease History	Dummy for been diagnosed with high blood pressure, diabetes, myocardial infarction, apoplexy, or bone fracture in the past	CHNS
Smoking	Dummy for smoking behavior	CHNS
Medical Insurance	Dummy for possession of any health insurance	CHNS
Abnormal BMI	Dummy for BMI larger than 24 or smaller than 18.5	CHNS
Trade Policy Shocks	Output tariff, elimination of export uncertainty, export quota policy, and export licenses	Fan et al. (2020)

Appendix Table 3 Summary Statistics of the CHNS Sample

Variables	Mean	SD	Min	Median	Max
Tariff	0.098	0.060	0.027	0.070	0.270
Illness or Injury	0.087	0.282	0.000	0.000	1.000
Muscle Pain or Fracture	0.062	0.241	0.000	0.000	1.000
Other Diseases	0.101	0.301	0.000	0.000	1.000
Log(Yearly Income(CHNS))	9.476	0.823	3.761	9.498	12.292
Log(Working Time)	3.845	0.282	2.303	3.871	4.836
Log(Hourly Wage)	1.847	0.755	-0.381	1.811	5.444
Female	0.335	0.472	0.000	0.000	1.000
Education Year	9.027	2.763	0.000	9.000	18.000
Age	37.212	10.645	16.000	38.000	59.000
Urban	0.368	0.482	0.000	0.000	1.000
Married	0.787	0.409	0.000	1.000	1.000
Minority	0.088	0.283	0.000	0.000	1.000
Disease History	0.066	0.248	0.000	0.000	1.000
Smoking	0.349	0.477	0.000	0.000	1.000
Abnormal BMI	0.343	0.475	0.000	0.000	1.000
Medical Insurance	0.609	0.488	0.000	1.000	1.000
$\text{Age}^{\text{Spouse}}$	39.326	8.259	18.000	40.000	59.000
$\text{Education Year}^{\text{Spouse}}$	8.659	3.236	0.000	9.000	18.000
$\text{Employment}^{\text{Spouse}}$	0.744	0.436	0.000	1.000	1.000
Tariff_Output	0.118	0.067	0.033	0.080	0.295
Export Policy Uncertainty	0.299	0.783	-0.543	0.000	3.297
Trade License Policy	0.018	1.077	-1.952	0.808	0.808
Quota Policy	0.407	1.128	-0.424	-0.023	4.098
Log(Air Quality Index)	4.466	0.404	3.151	4.482	5.260

Appendix Table 4 Summary Statistics of the UHS Sample

Variables	Mean	SD	Min	Median	Max
Tariff	0.096	0.036	0.023	0.088	0.198
Log(Yearly Income(UHS))	9.179	0.798	2.303	9.205	11.338
Female	0.437	0.496	0.000	0.000	1.000
Education Year	11.503	2.482	0.000	12.000	16.000
Age	40.993	8.562	16.000	41.000	59.000
$\text{Age}^{\text{Spouse}}$	41.387	12.097	16.000	42.000	95.000
Education $\text{Year}^{\text{Spouse}}$	10.350	3.538	0.000	12.000	16.000
$\text{Employment}^{\text{Spouse}}$	0.617	0.486	0.000	1.000	1.000
Tariff_Output	0.112	0.040	0.037	0.104	0.230
Export Policy Uncertainty	0.320	0.903	-0.669	0.000	4.656
Trade License Policy	-0.332	1.126	-2.106	-0.542	0.808
Quota Policy	0.213	1.312	-0.424	-0.389	9.915

Appendix Table 5 Initial Tariff and Contemporaneous Tariff

VARIABLES	(1) Full Sample	(2) CHNS City Sample	(3) UHS City Sample
Initial Tariff * 2000	-0.343***	-0.335***	-0.344***
	(0.006)	(0.019)	(0.007)
Initial Tariff * 2001	-0.439***		-0.444***
	(0.007)		(0.008)
Initial Tariff * 2002	-0.430***		-0.440***
	(0.007)		(0.008)
Initial Tariff * 2003	-0.528***		-0.531***
	(0.005)		(0.006)
Initial Tariff * 2004	-0.588***	-0.575***	-0.590***
	(0.006)	(0.013)	(0.007)
Initial Tariff * 2005	-0.602***		-0.605***
	(0.008)		(0.009)
Initial Tariff * 2006	-0.769***	-0.783***	-0.776***
	(0.005)	(0.011)	(0.007)
Initial Tariff * 2007	-0.761***		-0.769***
	(0.005)		(0.006)
Initial Tariff * 2008	-0.797***		-0.803***
	(0.005)		(0.006)
Initial Tariff * 2009	-0.775***	-0.791***	
	(0.005)	(0.012)	
Initial Tariff * 2010	-0.768***		
	(0.005)		
Initial Tariff * 2011	-0.773***	-0.792***	
	(0.005)	(0.012)	
Initial Tariff * 2012	-0.760***		
	(0.005)		
Initial Tariff * 2015	-0.802***	-0.823***	
	(0.005)	(0.012)	
Constant	0.174***	0.189***	0.178***
	(0.000)	(0.001)	(0.000)
Observations	5,160	364	2,450
R-squared	0.997	0.997	0.997
Year FE	Y	Y	Y
Prefecture FE	Y	Y	Y

Note: This table presents a simplified first-stage analysis at the prefecture-year level for the IV. We regress the current tariff levels on the interactions between the 1997 initial tariffs and wave dummies. Waves 2013 and 2014 are dropped due to data availability. Standard errors clustered at the prefecture-year level are in parentheses. ***, **, and * indicate significance levels of 1%, 5%, and 10%, respectively.